

Levered Noise and the Limits of Arbitrage Pricing: Implications for the Term Structure of Equity Risk Premia

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ABSTRACT

We show theoretically and empirically that no-arbitrage pricing magnifies the importance of noise when replication requires offsetting positions with similar fundamentals. This occurs because fundamentals are hedged, while any errors in the underlying asset prices are levered and amplified. Levered noise causes bias in estimated return moments including means, variances, and autocorrelations. We illustrate these effects in pairs trading strategies and in the highly levered portfolios required to replicate dividend strips. To improve estimates of the term structure of equity risk premia, we develop replication methods that reduce noise, focus on statistics that are more robust to noise, and provide time series that are longer than any prior study. The improved estimates provide no evidence of a downward sloping unconditional equity term structure. In the longest samples, dividend strip returns are lower than the market index.

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We show theoretically and empirically that no-arbitrage pricing magnifies the importance of noise when replication requires offsetting positions with similar fundamentals. This occurs because fundamentals are hedged, while any errors in the underlying asset prices are levered and amplified. Levered noise causes bias in estimated return moments including means, variances, and autocorrelations. We illustrate these effects in pairs trading strategies and in the highly levered portfolios required to replicate dividend strips. To improve estimates of the term structure of equity risk premia, we develop replication methods that reduce noise, focus on statistics that are more robust to noise, and provide time series that are longer than any prior study. The improved estimates provide no evidence of a downward sloping unconditional equity term structure. In the longest samples, dividend strip returns are lower than the market index.

I. Introduction

Ignoring frictions, absence of arbitrage allows us to obtain the value of a derivative from the prices of related assets. One well-known example decomposes a coupon-paying bond into a portfolio of zero-coupon strips, each of which may have a different discount rate. Other standard relationships include put-call parity and futures-spot parity, which respectively link the prices of puts and calls or forwards to a bond, the underlying asset, and the present value of dividends.¹

In this paper, we study no-arbitrage pricing when small errors are present in the observed prices of underlying assets. Price “noise” is typically defined as the deviation of observed price from fundamental value (also called the efficient price). Widely cited causes include bid-ask bounce, slow reaction to news, price pressure or order imbalances, and irrational traders and limits to arbitrage.² Noise affects inferences about return moments including simple mean returns, variances, covariances, and autocorrelations.³ Settings where noise is known to be empirically important are when returns are measured at relatively high daily or intraday frequencies, or at lower frequencies (e.g., monthly) in less liquid assets such as small stocks and low-attention stocks.

We show that even for nearly efficiently priced securities, noise can be economically important for derivatives pricing. This occurs because synthetic replication often requires considerable leverage, which magnifies noise. Concretely, assume a trading strategy where the fundamental value V can be decomposed into the true values of long (L) and short (S) positions. To accommodate noise, let observed value be:

$$\begin{aligned} V^o(L, S, \varepsilon_L, \varepsilon_S) &= Le^{\varepsilon_L} - Se^{\varepsilon_S} \\ &= L^o - S^o, \end{aligned} \tag{1}$$

¹More complex forms of arbitrage pricing include dynamic replication (Black and Scholes, 1973), asymptotic arbitrage and the arbitrage pricing theory (Ross, 1976), and statistical arbitrage such as pairs trading.

²Asparouhova, Bessembinder, and Kalcheva (2013) give a thorough discussion of the economics of noisy prices and prior literature. See also Hendershott and Menkveld (2014) and Brogaard, Nguyen, Putnins, and Wu (2019) for quantification of the magnitude of noise in daily stock returns.

³See, for example, Niederhoffer and Osborne (1966), Blume and Stambaugh (1983), Lo and MacKinlay (1990), Asparouhova, Bessembinder, and Kalcheva (2013), and Boguth, Carlson, Fisher, and Simutin (2016).

where the multiplicative noise terms satisfy $\mathbb{E}[e^{\varepsilon_L}] = \mathbb{E}[e^{\varepsilon_S}] = 1$. The elasticity of V^o with respect to its components is:⁴

$$d \ln V^o = d \ln L + \mathcal{L}(d \ln L - d \ln S) + d\varepsilon_L + \mathcal{L}(d\varepsilon_L - d\varepsilon_S), \quad (2)$$

where $\mathcal{L} = \frac{S^o}{L^o - S^o}$ captures leverage. Exposures to differences in fundamentals ($d \ln L - d \ln S$) and noise ($d\varepsilon_L - d\varepsilon_S$) become unbounded as S^o approaches L^o , i.e., as leverage increases. Hence, both differences in fundamentals and pricing errors are amplified.

In the common case where fundamentals on the long and short side are closely related, such as in a pairs trade, the changes in fundamentals offset ($d \ln L \approx d \ln S$). The second term on the right-hand side of (2) is then small. In this case, as leverage increases, the ratio of noise to fundamentals grows. Intuitively, arbitrage strategies are designed to hedge changes in fundamentals, but do not hedge asset-specific components, one of which may be noise.

Following prior literature (e.g., Blume and Stambaugh, 1983), once noise is large relative to fundamentals, bias in return moments becomes important. We provide analytical expressions for bias as a function of leverage. When price errors are independent and identically distributed (“iid”) across the long and short positions and time, net position return variance and simple mean returns are both biased upwards, and autocorrelations are biased downwards. These are the same effects originally identified by Blume and Stambaugh, but magnified by leverage.

When price errors are due to slow price adjustment, Lo and MacKinlay (1990) show that with positive portfolio weights portfolio variance is downward biased and autocorrelations are positive. Boguth, Carlson, Fisher, and Simutin (2016) further show that this causes downward bias in short horizon simple mean returns. We show that these results reverse in long-short portfolios. Intuitively, if one side of a long-short trade responds more quickly to fundamental news, the initial return does not fully incorporate the hedging effect of the opposing position, and the observed return will tend to reverse.

⁴This expression follows from totally differentiating the logarithm of equation (1): $d \ln V^o = \frac{\partial \ln V^o}{\partial \ln L} d \ln L - \frac{\partial \ln V^o}{\partial \ln S} d \ln S + \frac{\partial \ln V^o}{\partial \varepsilon_L} d\varepsilon_L - \frac{\partial \ln V^o}{\partial \varepsilon_S} d\varepsilon_S$.

Therefore, for the two primary types of noise considered in prior literature (iid noise and slow price adjustment), leverage in a long-short portfolio contributes to upward bias in mean returns and variances, and downward bias in autocorrelations.

We demonstrate the empirical validity of these predictions using pairs trading strategies. We go long $\mathcal{L} + 1$ shares on one side of a pair and short $\mathcal{L}$ shares of the other side, thereby holding exposure to fundamentals constant at one share while leveraging the exposure to noise. Consistent with our theoretical predictions, monthly simple return means and variances increase with the absolute value of $\mathcal{L}$. Effects on autocorrelations and betas are also consistent with predictions.

Understanding the biases caused by levered noise is particularly important for estimating the term structure of equity risk premia. The value of equity can be decomposed into portfolios of dividend strips, i.e. claims to dividends of different maturities, each of which may have a different discount rate (Brennan, 1997). The collection of these discount rates is the term structure of equity risk premia.

Binsbergen, Brandt, and Koijen (“BBK”, 2012) recognize the importance of the equity term structure for discriminating among asset pricing models. BBK replicate dividend strips using long-short positions in spot and derivative markets, relying on futures-spot and put-call parity relations. They find that returns on short-term dividend claims exceed those on long-term claims, suggesting a downward-sloping term structure. For example, the two dividend strip strategies they consider have mean excess returns of 88 and 84 basis points per month, more than three times larger than the S&P 500 index (27 basis points) in the same time period. (See their Table 1.) These results pose a challenge to traditional models of the equity premium, such as habit formation (Campbell and Cochrane, 1999) and long-run risk models (Bansal and Yaron, 2004), both of which imply unconditionally upward-sloping term structures, and disaster models (Barro, 2006), which imply a flat term structure. Spurred by the empirical finding of a downward-sloping term structure of equity risk premia, theorists have developed new models to match the proposed new empirical fact.⁵

⁵See, for example, Wang (2011), Croce, Lettau, and Ludvigson (2015), Belo, Collin-Dufresne, and Goldstein

Additional empirical evidence comes from proprietary “indicative prices” for dividend strips, obtained from individual financial institutions. Indicative prices are neither records of actual trades nor quotes representing offers to trade, but are maintained internally by financial institutions to support pricing of instruments such as over-the-counter (OTC) dividend swaps, and in some cases listed dividend futures. Binsbergen, Hueskes, Koijen, and Vrugt (2013) obtain an eight-year sample of indicative prices from BNP Paribas and Goldman Sachs, extended by three years in Binsbergen and Koijen (2017). They claim that these prices support the conclusion of a downward-sloping term structure, but the evidence is mixed across markets, time periods, and methodologies.⁶ Bansal, Miller, Song, and Yaron (2018) investigate similar indicative prices from an unnamed financial institution, and in a similar time period come to the opposite conclusion of a downward-sloping unconditional term structure.⁷ They also obtain the limited available data on listed dividend futures and emphasize low trading volume and high trading costs. Mixon and Onur (2017) obtain comprehensive regulatory data from the Commodity Futures and Trading Commission on actual trades in OTC dividend swaps. They document very low activity, and urge caution in interpreting empirical findings related to these markets. In sum, indicative, listed, and OTC dividend swap and futures prices suffer from short samples, issues of data quality and availability, and in combination with a variety of methodologies have led to mixed empirical results.

We reevaluate the initial, seemingly very powerful, evidence of a downward-sloping term structure in synthetic dividend strips. Our theory of levered noise immediately calls into question the existing evidence, but also more positively suggests improved empirical methods. Matching our theory, replicating a dividend strip requires high leverage with

(2015), and Hasler and Marfè (2016).

⁶The evidence in Binsbergen, Hueskes, Koijen, and Vrugt (2013) is largely based on dividend “yields,” which are not actual returns, but rather estimates of expected return that require as inputs estimates of dividend growth rates. These are obtained by a VAR on their eight-year time series. The resulting yields vary considerably across markets and time. See their Figure 6. Binsbergen and Koijen (2017) calculate zero-cost profits to trades on indicative futures prices and add these to bond returns to compare across maturities and to a market index. The results again vary across markets, with comparisons across maturities sometimes suggesting a different slope than comparisons between a given maturity and the index. See their Figure 2.

⁷Bansal, Miller, Song, and Yaron (2018) provide additional evidence on the *conditional* term structure that is consistent with some traditional asset pricing models including long-run risk. This conditional evidence is qualitatively similar to findings in Binsbergen, Hueskes, Koijen, and Vrugt (2013).

offsetting positions that have similar fundamentals. Specifically, one simple replication is to buy an index claim and short an offsetting one-year futures. The net claim entitles one to all dividends over the next year. Its price is a fraction of the value of the long or the short side of the trade. Quantitatively, assume an annualized dividend yield of 2%, roughly consistent with prior experience.⁸ Normalizing the long value to \$100, the offsetting futures position would have a notional amount in the neighborhood of \$98, and the strip value would be about \$2. Consider small microstructure frictions of a few basis points (cents) on either the long side or the short side, or on the synchronicity of the two prices. This mispricing may be irrelevant compared to the value of the long side, but substantial relative to the net value of the strip.

BBK report not only high means of monthly dividend strip returns, but also high volatility, low beta, strong negative autocorrelations (on the order of -30%), and time series predictability. All of these are consistent with our theory of levered noise.

We identify a timing mismatch as one specific source of noise in the BBK methodology. They sample intraday option prices between 10AM and 2PM, while using interest rates largely determined by Eurodollar futures settlement prices from the Chicago Board Option Exchange (CBOE), at 3PM ET.⁹ The timing mismatch is important: We show that 15% of the variation in the BBK return is predicted by the interest rate change on the final day of the preceding month.

We fix this problem by developing alternative methodologies based on end-of-day prices from OptionMetrics. This significantly reduces the timing mismatch between option quotes and interest rate observations, which we demonstrate by showing that predictability of returns from prior interest rate changes is much smaller.¹⁰ Consistent with lower noise, our returns have lower variance and less negative autocorrelations

⁸During the 1996-2009 sample period considered by BBK, the U.S. annualized price-dividend ratio averaged approximately 60, ranging from a low of 28 to a high of 90, where the monthly stock market price-dividend ratio is calculated following the method of Shiller (2005) with data downloaded from <http://www.econ.yale.edu/shiller/data.htm>.

⁹They do not specify the time zone, which could be the Central Time of the option markets, or the Eastern Time of the underlying index.

¹⁰Another important advantage of our approach is that the OptionMetrics data is widely used in finance research and much less expensive than the primary data source in the BBK methodology. Our results are therefore straightforward for other researchers to replicate.

than the BBK returns.

Nonetheless, our estimates of strip prices remain noisy. To diagnose this we develop generic measures of noise, based on the cross-sectional dispersion in candidate strip prices calculated at the same point in time from different strikes. These measures significantly predict future reversals, and controlling for noise significantly reduces simple mean returns.

Finally, we use mean logarithmic returns, which are less biased by noise than mean simple returns, to evaluate the relative returns to the different dividend strip strategies and the market. In the BBK sample period, all of our OptionMetrics-based estimates of dividend strip returns are lower than the BBK estimates, as well as less noisy. Further, the OptionMetrics-based dividend strip returns are similar to or lower than market returns in the same period. We also extend the sample outside the BBK sample period, which ends in October 2009, to April 2016. In the extended sample, all of the estimated mean logarithmic returns for dividend strips are lower than for the market. In short, with either improved methodology or an extended sample, and especially with both, there is no evidence of a downward-sloping term structure in equity risk premia.

Our paper is one of several that have begun to question the orthodoxy of a downward-sloping term-structure.¹¹ Schulz (2016) shows that differential tax treatment between dividends and capital gains can cause the relative pre-tax returns of high-yielding dividend strips versus the underlying index to be a poor reflection of their relative after-tax returns. Song (2018) provides evidence that mismeasurement of dealer funding costs is an important source of noise in estimated synthetic strip prices and that this could lead to overestimation of dividend strip returns. As mentioned previously, Mixon and Onur (2017) recommend caution in interpreting empirical results based on indicative prices

¹¹Our earlier working paper, (Boguth, Carlson, Fisher, and Simutin, 2012), first made the point that levered noise *could* cause significant bias in reported dividend strip mean returns and other moments, that the magnitudes of the biases could be large, and that circumstantial evidence supported this view since a unified model of noise could explain all of the reportedly anomalous facts. Our current manuscript adds to this work. We (i) show dividend strip returns for a broader range of synthetic replication strategies including new methods that reduce noise and use data more accessible to other researchers; (ii) provide direct evidence of the noise in different replication strategies allowing comparisons; and (iii) study an extended sample with much longer time series than existing literature.

relating to OTC dividend swap contracts, because of very low trading activity. Bansal, Miller, Song, and Yaron (2018) similarly find high trading costs, low activity, and short time spans for listed dividend futures contracts, and their examination of indicative prices from an unnamed financial institution reveals an upward sloping unconditional term structure. We add to these studies by demonstrating that the initial findings of high synthetic dividend strip returns were biased, and by developing improved empirical methods and longer time series. These more accurate and less biased point estimates of dividend strip returns are less than those of the market index.

More broadly, our paper relates to the literature on limits to arbitrage, which emphasizes that arbitrageurs may fail to bring prices fully toward their fundamental values for a variety of reasons.¹² The point we make is different, relating to the implications of arbitrage opportunities for noise in prices, and the leverage required to execute an individual arbitrage trade.

II. Noise in Levered Portfolios

Earlier literature gives examples of noise in prices, including bid-ask bounce and slow price adjustment (e.g., Blume and Stambaugh, 1983, Lo and MacKinlay, 1990). Nonetheless, it is worth giving our views on how to interpret “noise” in the context of our theory and empirical work.

Noise is always defined relative to a model. From the point of view of a microstructure theorist, the reasons for slow price adjustment might be the central focus of a model, and are exactly the features of the data to be explained, not noise. Many leading asset pricing models, however, are not designed to capture small microstructure frictions such as bid-ask bounce, slow price adjustment, or small violations of the law of one price. Generally, consumption-based models such as habit formation (Campbell

¹²These reasons include the leverage and solvency of arbitrageurs, the separation of capital from knowledge of mispricing, uncertainty over how long mispricing can persist, short-sales constraints, and fire-sale externalities. See, e.g., Miller (1977), Shleifer (1986), Shleifer and Summers (1990), De Long, Shleifer, Summers, and Waldmann (1990), Shleifer and Vishny (1997), Kyle and Xiong (2001), Gromb and Vayanos (2002), Morris and Shin (2004), Allen and Gale (2005), Pontiff (2006), Brunnermeier and Pedersen (2009), Stein (2009), Frazzini and Pedersen (2014).

and Cochrane, 1999), long-run risk (Bansal and Yaron, 2004), or rare disasters (Barro, 2006), knowingly ignore small frictions in prices. From the point of view of these models, microstructure phenomena are noise, and we would not want the impacts of small price frictions to influence our judgments about the validity of these theories.

Following prior notation, let V_t be the fundamental value of a claim at time t and let the observed price be:

$$V_t^o = V_t e^{\varepsilon_t}, \quad (3)$$

where ε_t represents “noise” and satisfies $E[e^{\varepsilon_t}] = 1$. The single-period gross simple return is

$$R_{t,1} = \frac{V_{t+1} + D_{t+1}}{V_t}. \quad (4)$$

Applying a logarithmic transform and linearizing following Campbell and Shiller (1988) gives:

$$r_{t,1} \approx k + \rho(v_{t+1} - v_t) + (1 - \rho)(d_{t+1} - v_t), \quad (5)$$

where $r_{t,1} = \log(R_{t,1})$, $v_t = \log(V_t)$, $d_t = \log(D_t)$, and ρ and k are constants whose values are given in the Appendix. This equation links returns to fundamentals: one-period log returns are a weighted average of the change in log value and the realized log cash-flow yield.

Aggregation in time leads to:

$$\begin{aligned} r_{t,T} &= \sum_{\tau=1}^T r_{t+\tau-1,1} \\ &= Tk + \rho(v_{t+T} - v_t) + (1 - \rho) \sum_{\tau=1}^T (d_{t+\tau} - v_{t+\tau-1}). \end{aligned} \quad (6)$$

The logarithmic long-horizon return depends on the beginning and ending security values and on the sum of all intermediate realized dividend yields.

The log-linearization easily accommodates the multiplicative errors:

$$\begin{aligned} r_{t,T}^o &= \sum_{\tau=1}^T r_{t+\tau-1,1}^o \\ &= r_{t,T} + \rho(\varepsilon_{t+T} - \varepsilon_t) - (1 - \rho) \sum_{\tau=1}^T \varepsilon_{t+\tau-1}. \end{aligned} \quad (7)$$

The logarithmic long-horizon observed return is the true return plus two noise terms. The first noise term, $\rho(\varepsilon_{t+T} - \varepsilon_t)$, is well understood and reflects the difference between the final and initial errors: all intermediate errors in the ex-dividend component of returns cancel out. The impact of this term diminishes quickly as return horizon increases.

The final term in Equation (7) is new to the literature and reflects that noise-free dividends are reinvested at noisy observed prices. Intermediate errors therefore do not cancel out. Prior literature, e.g., Blume and Stambaugh (1983), Lo and MacKinlay (1990), considers only non-dividend paying assets and therefore does not have this channel. Understanding the effects of noise in the presence of dividend reinvestment may be important for high-yielding assets, such as some dividend strips.¹³

A. Levered Noise

We show the effects of noise on returns for long-short strategies of the form:

$$V_t^o = (\mathcal{L} + 1)L_t^o - \mathcal{L}S_t^o \quad (8)$$

with dividends given by

$$D_t = (\mathcal{L} + 1)D_{L,t} - \mathcal{L}D_{S,t}. \quad (9)$$

We consider applications in which values and dividends of the long and short positions are cointegrated. We can then log-linearize the prices and dividends of these long-short portfolios:¹⁴

$$v_t^o = l_t + \mathcal{L}(l_t - s_t) + \varepsilon_{L,t} + \mathcal{L}(\varepsilon_{L,t} - \varepsilon_{S,t}) \quad (10)$$

and

$$d_t = d_{L,t} + \mathcal{L}(d_{L,t} - d_{S,t}). \quad (11)$$

Denoting the logarithmic total return of the trading strategy over the horizon $[t, t + T]$ by $r_{t,T}^o$, equation (7) implies that

$$r_{t,T}^o = r_{L,t,T} + \mathcal{L}(r_{L,t,T} - r_{S,t,T}) + \xi_{L,t,T} + \mathcal{L}(\xi_{L,t,T} - \xi_{S,t,T}), \quad (12)$$

¹³The noise-free dividend in Equation (7) can also be interpreted as any other certain payment, such as a bond coupon, or in the case of rollover strategies, a bond face value payment.

¹⁴The details are provided in the Appendix.

where

$$\xi_{i,t,T} = \rho(\varepsilon_{i,t+T} - \varepsilon_{i,t}) - (1 - \rho) \sum_{\tau=1}^T \varepsilon_{i,t+\tau-1}, \quad (13)$$

for $i = L, S$ is the total noise in the T -period return on each leg of the trade. The final term of Equation (12) shows that leverage $\mathcal{L}$ magnifies the importance of noise in returns.

B. Biases in Levered Return Moments

We show the effects of noise on levered portfolio return moments. The moments we consider are the mean logarithmic return, and the mean, variance, autocovariance, covariance and beta of simple returns.

Prior literature, e.g., Blume and Stambaugh (1983), Lo and MacKinlay (1990) concludes that expected logarithmic returns are not affected by noise, based on analysis of non-dividend paying portfolios. We take expectations of both sides of Equation (7) to analyze impacts on the mean logarithmic returns of dividend-paying assets. By Jensen's inequality the standard assumption that observed prices provide unbiased estimates of true prices, i.e., $E[e^{\varepsilon_t}] = 1$, implies that $\mathbb{E}(\varepsilon_t) < 0$. When the dividend yield is positive, i.e., $1 - \rho > 0$, we obtain $E(r_{t,T}^o) > E(r_{L,t,T})$.¹⁵ The upward bias in mean logarithmic returns is larger when price noise is larger and when the dividend yield is higher.

A rough calibration, using moments of returns calculated later in the paper, convinces us that the bias in mean logarithmic returns from dividend reinvestment will not be large enough to meaningfully impact the strategies used in our paper.¹⁶ This conclusion does not mean that all of the mean logarithmic returns calculated in our paper are reliable. In particular, Schulz (2016) shows that some dividend strip strategies collect a large fraction of their value in dividends, which have a tax disadvantage

¹⁵For example, in the Gaussian case, $\varepsilon_t \sim \mathcal{N}(-1/2\sigma_\varepsilon^2, \sigma_\varepsilon^2)$, and $\mathbb{E}(r_{t,T}^o) = \mathbb{E}(r_{t,T}) + T/2(1 - \rho)\sigma_\varepsilon^2$.

¹⁶Our calibration focuses on an 18-month dividend strip, approximating the BBK "short-term asset" studied later in the paper. The average monthly dividend yield is on the order of 5.5%, and we take as a coarse approximation an iid noise volatility of $\sigma_\varepsilon = 4.5\%$. This produces an upward bias in the mean logarithmic return of less than one basis point per month. To obtain the noise volatility, we approximate the monthly dividend strip true return volatility to be equal to that of the index (about 5%), which implies an iid noise standard deviation of about 4.5% to produce the total volatility of about 8% monthly. This calibration also matches the short-term asset autocorrelation of approximately -30% in monthly returns. See Table 1 for the moments used in the calibration.

relative to other forms of income. In equilibrium the pre-tax returns of these strategies must include compensation for this tax disadvantage. This bias in returns affects dividend strips that collect dividends, but not others that avoid dividends by trading only futures or synthetic futures, as described in Section IV.

We now show that for both dividend-paying and non-dividend-paying assets, a variety of moments of simple mean returns are severely biased in the presence of levered noise. Our analysis focuses on the two types of noise that have been the focus of prior literature on measurement issues in prices: independent errors (Blume and Stambaugh, 1983) and slow price adjustment (Lo and MacKinlay, 1990). We provide expressions for biases in return mean, variance, autocovariance, and beta in these two scenarios.

B.1. Independent Measurement Error

For simplicity, we assume that the long and short sides of Equation (8) have identical fundamentals, and that their noises are independent of one another. Taking the variance of Equation (12) gives the variance of logarithmic T -period returns. The observed variance is biased upward by noise:

$$\text{Var}(r_{t,T}^o) = \sigma_{L,T}^2 + (\mathcal{L} + 1)^2 \sigma_{\xi_{L,T}}^2 + \mathcal{L}^2 \sigma_{\xi_{S,T}}^2 > \sigma_{L,T}^2, \quad (14)$$

where $\sigma_{L,T}^2 = \text{Var}(r_{L,t,T})$ is the variance of fundamental returns and $\sigma_{\xi_{i,T}}^2$ is the variance in noise of the T -period return on leg $i = L, S$ of the trade. In the special case where the noises are also independent across time and identically distributed, as in Blume and Stambaugh (1983), the noise variance at horizon T is

$$\sigma_{\xi_{i,T}}^2 = [2\rho + T(1 - \rho)^2] \sigma_{\varepsilon_i}^2. \quad (15)$$

In absence of leverage ($\mathcal{L} = 0$) and dividends ($\rho = 1$), the return variance is inflated by double the variance of iid noise, as in Blume and Stambaugh (1983). Leverage magnifies the bias, as shown by the quadratic dependence on $\mathcal{L}$. With dividends ($\rho < 1$) the impact of noise increases with T , implying that bias can remain important at longer horizons than in the absence of dividends.

Noise has implications for other moments of simple returns. Following Jensen's inequality, bias in variance causes bias in mean simple returns:

$$\mathbb{E}(R_{t,T}^o) \approx e^{E(r_{t,T}^o) + \text{Var}(r_{t,T}^o)/2} > \mathbb{E}(R_{t,T}) \approx e^{E(r_{t,T}) + \text{Var}(r_{t,T})/2}. \quad (16)$$

Also, consistent with excess volatility (Shiller, 1981), the observed short-term asset return is predictable with first-order autocovariance

$$\text{Cov}(r_{t+T,T}^o, r_{t,T}^o) = \text{Cov}(r_{t+T,T}, r_{t,T}) - \rho [(\mathcal{L} + 1)^2 \sigma_{\varepsilon_L}^2 + \mathcal{L}^2 \sigma_{\varepsilon_S}^2]. \quad (17)$$

Measurement error unambiguously biases the measured autocovariance downwards. Leverage again inflates the importance of noise by factors on the order of $\mathcal{L}^2$.

The covariance with respect to the observed market is straightforward to compute from equation (12). If the fundamentals in the long and short sides perfectly hedge and the noises are uncorrelated with the market, then beta should be flat as leverage varies. If the long and short sides do not perfectly hedge or the long-short noise is correlated with the market, then covariance and beta vary linearly with leverage.

B.2. Lagged Price Adjustment

We now consider errors in pricing induced by the delayed reaction of prices to information. The analysis is similar in spirit to Lo and MacKinlay (1990), but we consider the effects on long-short portfolios, as opposed to long-only, and demonstrate the effects of leverage. For simplicity, we assume that neither asset pays dividends and the fundamentals of the long and short positions are identical, following:

$$l_{t+1} = s_{t+1} = l_t + \mu - 1/2\sigma^2 + \sigma z_{t+1}, \quad (18)$$

where μ and σ are the mean and standard deviation of one-period log returns, and z_{t+1} is an independent $N(0, 1)$ shock. For notational simplicity, we further restrict delayed price reaction to the long leg of the trade. To model delayed reaction to fundamentals, a state variable η_t captures immediate underreaction to contemporaneous news, which then gradually diffuses into future prices at an exponential rate:

$$\eta_{t+1} = -\lambda \sigma z_{t+1} + e^{-\kappa} \eta_t, \quad (19)$$

where $\kappa > 0$ and $0 \leq \lambda < 1$. The observed log price of the long position is $l_t^o = l_t + \eta_t$.

The logarithmic return of the net position is

$$r_{t,1}^o = \mu + \sigma z_{t+1} + (\mathcal{L} + 1) (\eta_{t+1} - \eta_t) \quad (20)$$

$$= \mu + [1 - \lambda(\mathcal{L} + 1)] \sigma z_{t+1} + (\mathcal{L} + 1) (e^{-\kappa} - 1) \eta_t. \quad (21)$$

The long-short strategy return variance and autocovariance are:

$$\text{Var}(r_{t,1}^o) = \sigma^2 \left[1 - 2\lambda(\mathcal{L} + 1) + \frac{2}{1 + e^{-\kappa}} \lambda^2 (\mathcal{L} + 1)^2 \right] \quad (22)$$

$$\text{Cov}(r_{t,1}^o, r_{t+1,1}^o) = \sigma^2 \left[(1 - e^{-\kappa}) \lambda (\mathcal{L} + 1) - \frac{1 - e^{-\kappa}}{1 + e^{-\kappa}} \lambda^2 (\mathcal{L} + 1)^2 \right] \quad (23)$$

Slow price adjustment increases variance and causes negative autocorrelations when leverage is sufficiently large.

The observed beta with respect to the market is easily understood by examining equation (20). If the fundamental z_t is positively correlated with the market, then slow reaction causes contemporaneous beta to be biased downward, and beta with respect to market lags would be positive. If instead the slow reaction is on the short side, then the contemporaneous beta would be upward biased, and lagged betas would be negative. In both cases, using an approach such as a “sum” beta (Dimson, 1979) would give an accurate estimation of the true loading on the market.

III. Levered Noise in Equity Pairs Trades

We illustrate the theoretical predictions of Section II regarding the impact of long-short trade leverage on return moments using several well known equity pairs trades. We choose three examples from among a large literature.¹⁷ First, we examine a simple example of cross-listings: shares of the Royal Bank of Canada that are traded on both the Toronto Stock Exchange and the New York Stock Exchange. Since exactly the same

¹⁷Examples of equity arbitrage include cross listing, i.e., listing of the same or similar shares on different exchanges (Gagnon and Karolyi, 2010), with the well-know case of Royal Dutch Shell (De Jong, Rosenthal, and Van Dijk, 2009), dual class shares involving different cash flow, voting, or conversion rights (Schultz and Shive, 2010), merger arbitrage (e.g., Mitchell and Pulvino, 2001, Baker and Savasoglu, 2002), and stub trades in subsidiaries and parent firms (Mitchell, Pulvino, and Stafford, 2002), and more general pairs trades of price pairs that have historically comoved closely (Gatev, Goetzmann, and Rouwenhorst, 2006).

shares are traded on both exchanges, the only difference is due to exchange rates, and we expect tight adherence to no-arbitrage pricing. Second, we consider Berkshire Hathaway A and B shares, which have proportional cash-flow rights. The more expensive A shares are convertible into B shares, but not vice versa.¹⁸ Finally, we consider a classic pairs-trading example from the literature: Royal Dutch Shell shares traded on the London and Amsterdam exchanges. These shares have claims to proportional cash flow rights but the dividends paid need not be identical at any point in time and taxation may be different. Hence we expect greater potential for deviations from long-run parity pricing for this trade.

The objective of standard equity arbitrage is to trade directionally when violations of a parity relation occur. Our purpose is different, and we seek simply to illustrate the theory of Section II, varying leverage while holding net exposure to one side of the trade constant. In particular, we go long $\mathcal{L} + 1$ shares of one side of the trade and short $\mathcal{L}$ shares of its pair.¹⁹

Figure 1 summarizes the relationship between leverage, stated as the number of shares $\mathcal{L}$ held short, and return moments for the case of a cross listing that is long Royal Bank of Canada shares traded on the New York Stock Exchange and short shares traded on the Toronto Stock Exchange and converted to US dollars. Panel A shows that the price deviations are small, typically much less than one percent of the value of the share, and short-lived. Panel B shows the relationship between leverage and average simple and logarithmic monthly returns. The graph shows that average logarithmic returns are approximately constant for the range of leverage considered.²⁰ In contrast,

¹⁸The ratio of A to B share cash-flow rights was traditionally 30:1, but grew to 1500:1 following a stock split of the B shares in 2010. The ratio of voting rights exceed the ratio of cash flow rights.

¹⁹Although such strategies are not explicitly studied in prior literature, following Mitchell, Pulvino, and Stafford (2002) it is common practice to consider the returns on arbitrage trades capitalized with initial margin proportional to the value of the long and short positions (e.g., one-half the value of the long and short positions is placed in a treasury security). In the trades we examine, the margin at every instant is one long share and, as a consequence, capital as a fraction of the (cointegrated) $\mathcal{L}$ -share long/short positions is constant through the life of the trade. Analyzing the impact of leverage in trades where treasuries are posted as margin capital would necessitate specifying maintenance margin regulations with associated margin calls as well as tracking interim purchases and sales of treasuries.

²⁰The small negative slope is caused by exchange-rate-adjusted US shares declining very slightly relative to the Canadian shares over the approximately 20-year investing horizon.

the simple returns increase with leverage, whether positive or negative. Consistent with equations (14) and (16), the difference between average logarithmic and simple monthly returns can be attributed to the quadratic relationship between leverage and return standard deviation displayed in Panel C. Following our theory, these U-shaped patterns in simple means and standard deviations reflect levered noise.

Panel D gives evidence of this noise. While US shares ($\mathcal{L} = 0$) have a slightly positive autocorrelation, consistent with the predictions in equations (17) and (23) the autocorrelation turns strongly negative with leverage. Panel E shows that beta increases as the number of long shares increases, consistent with US trades incorporating US market information faster than Canadian trades.

Panel F shows plots of variance ratios

$$VR_T = \frac{\text{Var}(r_{t,T}^o)/T}{\text{Var}(r_{t,252}^o)/252},$$

which are scaled variances at different horizons. Variance ratios are commonly used to summarize information in autocorrelations at multiple horizons (e.g., Lo and MacKinlay (1990)). Independent returns imply flat variance ratios, while negative autocorrelations produce downward-sloping variance ratios. We show the variance ratio plots for different degrees of leverage. When leverage is higher, the variance ratios are larger for small T , but in all cases the variance ratios decay fairly quickly and are close to one after two to three months. This indicates that the departures from no-arbitrage are relatively short-lived.

Figures 2 and 3 repeat this analysis for Berkshire Hathaway and for Royal Dutch Shell. For both Figures, Panel A shows that deviations from parity are more persistent than in the case of Royal Bank. For Berkshire Hathaway the Class B shares trade at a discount during times of stress (most notably during the dot-com crash in 2000 and the financial crisis in 2008). Royal Dutch Shell discrepancies are extremely long-lived, a puzzle that dates back to at least Rosenthal and Young (1990). From Panel B of both Figures, mean logarithmic returns are relatively unaffected by leverage. However, mean simple returns and variances show a U-shaped pattern in leverage, again consistent with

levered noise. Panel D of both Figures shows negative autocorrelations for all leverage levels. From Panel E of both Figures, betas are downward sloping in leverage, suggesting differences in the speed of response to new information across the long and short legs as in the discussion of Section II. The beta results confirm that the U-shaped patterns in mean simple returns are not related to leverage in systematic risk exposures.

Figures 2 and 3, Panel F, show strong horizon effects in variance ratios. In both cases, variances remain inflated even at longer horizons, one quarter to one year, when leverage is high.

A natural question to ask is whether the high simple mean returns achieved for both high and low leverage levels represent real profit opportunities. We tend to say no. First, recorded prices may not be tradeable. Second, the act of trading may move the market price. Third, the high negative autocorrelations in the returns imply that any profits that could be generated disappear over longer holding horizons. Fourth, we do not account for trading costs.

On the other hand, one might argue that our examples must be indicative of a profit opportunity somewhere. The law of one price is apparently being violated, if only to a small degree. It may be possible that some types of traders, particularly liquidity providers, could trade at advantageous prices closer to those observed in the data. These types of traders could potentially profit from noise, such as differences between bids and asks or small deviations of prices from fundamentals due to liquidity impacts, as argued in Asparouhova, Bessembinder, and Kalcheva (2013).

Regardless of one's thoughts about the potential profit opportunities for high-frequency traders, it is clear from the examples that these high simple mean returns tell us nothing about the risks in standard consumption-based asset pricing models. Since we use exactly or nearly identical assets on both sides of the trade, and create synthetic leverage, our net position always has the same exposure to "fundamentals." The high mean returns relate to price deviations that from the point of view of consumption-based asset pricing are noise.

IV. Dividend Strips and the Term Structure of Equity Risk Premia

Dividend strips entitle the holder to all dividends paid between dates $t + T_1$ and $t + T_2$, where t denotes the current date. Futures contracts are not dividend protected, thus ignoring frictions strip prices must be given by

$$V_{t,T_1,T_2} = e^{-r_{t,T_1}T_1}F_{t,T_1} - e^{-r_{t,T_2}T_2}F_{t,T_2}, \quad (24)$$

where $F_{t,T}$ is the forward price and $r_{t,T}$ is the risk-free rate of interest for payments with maturity $t + T$. Application of put-call parity allows rewriting (24) as

$$V_{t,T_1,T_2} = C_{t,T_1} - P_{t,T_1} + X_{T_1}e^{-r_{t,T_1}T_1} - C_{t,T_2} + P_{t,T_2} - X_{T_2}e^{-r_{t,T_2}T_2}, \quad (25)$$

where $P_{t,T}$ ($C_{t,T}$) is the price of a put (call) with maturity $t + T$ and exercise price X_T .

When $T_1 = 0$ the long side of the strip reduces to the market index. The strategy then collects dividends and has a positive near-term dividend yield ($1 - \rho > 0$), which can be very high. For example, in an eighteen-month strip, the dividend yield exceeds five percent per month, and 50% per year. Dividends compose a very high fraction of the total return. Schulz (2016) shows that because dividends are taxed at a higher rate than other forms of income, the pre-tax returns on dividend-collecting dividend strips are not immediately comparable with other assets. In particular, the returns on dividend-collecting dividend strips must be higher to compensate for their higher effective tax rates. This effect is not a reflection of net returns to investors or risk, but differential taxation. BBK point out the additional difficulty that dividend-collecting strategies require exact index replication, which is not costless. For these reasons even the logarithmic returns of dividend-collecting dividend strips are biased upward.

When $T_1 > 0$, the long side of the strip is a futures or synthetic futures. The strategy does not collect dividends in the short-run, and has a zero near-term dividend yield ($1 - \rho = 0$). When appropriately rolled over, these strategies, including one developed by BBK and several that we contribute, need not ever collect dividends and are therefore not impacted by the tax effect pointed out by Schulz (2016). Whenever

we make inferences about the term-structure of equity risk premia, we rely on the logarithmic returns of zero-yield strips.

We estimate prices of synthetic dividend strips by substituting bond prices and some combination of futures, option, and index values into equations (24) or (25). Small amounts of noise from any one of these primary markets, or imperfections in the synchronization of measured prices across the markets, can become levered in the synthetic prices and returns. Because the strips are based on relatively small time intervals $[T_1, T_2]$, the leverage implicit in replication is large. Further, the long and short sides of the trade will have very similar exposures to fundamentals. These facts fit our theory well, suggesting considerable potential for levered noise to impact return moments.

A. Options, Futures, and Interest Rate Data

S&P 500 index futures trade on the Chicago Mercantile Exchange (CME), and have maturities at three month intervals. Trading hours end at 4:15PM ET, 15 minutes later than the close of US stock markets. We obtain the prices of all intraday futures trades from 1996 to 2014 from TickData Inc. The “on the run” nearest-to-maturity contracts are highly liquid, but contracts with longer maturities are very thinly traded.²¹

S&P 500 index options trade on the Chicago Board Options Exchange (CBOE), which closes at the same time as the CME. We use two data sources for these options. First, we use end-of-day bid and ask quotes from OptionMetrics’ IvyDB, with our sample beginning in January 1996 and ending in April 2016. This data is widely used in academic research and available on Wharton Research Data Services.

Accurate time stamps are crucial to reducing noise in synthetic replication. According to the “IvyDB US File and Data Reference Manual” (2017), “OptionMetrics compiles the IvyDB data from raw 3:59PM EST price information.” However older

²¹In month-end trade days during the time periods we investigate, off-the-run futures contribute less than three percent of total trade volume. BBK (p. 1602) choose not to report results based on futures data due to the lack of available data for longer maturities, but state in their pre-publication drafts that their main findings are robust to using futures data. We use the same data source for futures data that they report.

versions of the manual are less clear about an exact time stamp, and simply refer to end-of-day quotes. In order to accurately identify end-of-day timestamps in the OptionMetrics data, we apply put-call parity to produce a daily time series of synthetic nearest to maturity futures prices and compare these options-based prices to the related futures prices extracted from TickData with timestamps that are nearest but prior to 4:00PM ET and 4:15PM ET. For every month, we compute the standard deviation of the daily synthetic/actual futures spread. The time series of the difference in these standard deviations clearly identify a structural break in March 2008. We infer that prior to March 2008, OptionMetrics used prices from the end of the CBOE trading day, at 4:15PM ET. After March 2008, OptionMetrics switched to the 3:59PM ET standard in current use, presumably to better match closing stock market prices.²²

We also use subsamples of the CBOE MDR data, which is the primary data source used by BBK, but expensive and otherwise not widely used by finance academics.²³ This database contains intraday time-stamped quotes for the same option contracts as in OptionMetrics, as well as time-matched reported values of the underlying index.

Trade volume in long-maturity S&P 500 option contracts is low in comparison with quote activity. We follow BBK by basing all of our S&P 500 option “prices” on quotes. Specifically, we take price, whether end-of-day (OptionMetrics) or intraday (MDR), as the midpoint of the reported bid and ask.

There are many types of S&P 500 index options. We deviate from BBK by selecting for our study only contract types with the longest history and the highest liquidity. Specifically, for both the MDR and OptionMetrics data, we restrict our samples to European options with expiration on the third Friday in the March-June-September-December cycle (June-December for maturities greater than one year), morning set-

²²Using our MDR subsamples of intraday quotes described below we confirm that the end-of-day OptionMetrics quotes before March 2008 match approximately to 4:15 timestamps, and after March 2008 match approximately to 4:00 timestamps.

²³The data is available at a price of USD 200 per month, or approximately USD 50,000 for the entire sample of data available from January 1996. Our MDR subsamples are selected from months within the span of the BBK sample period: 01/1996-07/1996, 10/2000-10/2001, and 01/2009-07/2009. These samples include seven months each from the first and last years of the BBK sample, as well as an interim period associated with high dividend strip return volatility.

tlement, and SPX as the underlying index (secid is 108105). This restriction has a marginal impact in helping to reduce the measures of noise that we report, but does not fundamentally alter our results.

Following BBK, the bond yields that we use for discounting are from OptionMetrics. Short-maturity yields (typically up to one month) are from LIBOR fixings (11:00AM GMT, 5:00AM ET), while longer maturity yields utilize daily settlement prices of CME Eurodollar futures (3:00PM ET).²⁴ Interest rates for specific investing horizons are linearly interpolated.

We also obtain two monthly dividend strip return series that BBK have made publicly available, which they constructed from MDR data, from February 1996 to October 2009. We use their time series directly whenever investigating their results.

B. BBK Strategies: “Short-term Asset” and “Steepener”

BBK consider two dividend strip strategies. Their “short-term-asset” is long the index ($T_1 = 0$) and short a synthetic discounted futures position, the price constructed from options quotes, with T_2 between 1.3 and 1.9 years. Much of the total return of the short-term asset comes from dividends. Following our prior discussion of high-yielding dividend strips, high dividend taxation (Schulz, 2016) and the cost of instantaneously tracking the index (BBK) cause the logarithmic returns of the short-term asset to be upward-biased and not reliable for inference about the equity term structure.

The BBK “steepener” uses the same short position as the short-term-asset but sets T_1 as close as possible to one year prior to T_2 . The long side is represented by a synthetic futures. This strategy does not collect dividends, and therefore dividend taxation does not cause an upward bias in its mean logarithmic return.

Option maturities greater than one year have June or December expiration dates, meaning that both of the BBK dividend strip strategies require rolls into new contracts on the short side every six months. The steepener strategy additionally requires

²⁴See the OptionMetrics IvyDB US Reference Manual available at WRDS and the CME settlement details at <http://www.cmegroup.com/market-data/settlements/settlements-details.html>.

similar turnover in the long leg, whereas the long side of the short-term asset consists of simply holding the underlying index. These facts suggest potentially larger noise in the steepener, even though the short-term asset has the significant disadvantage of upward-biased mean logarithmic returns due to taxation.

BBK provide monthly returns for their short-term asset and steepener. To obtain daily estimates of prices, as in Equation (25), in our MDR subsamples, we apply the following algorithm which is based on the description in BBK. Every call price quoted between 10:00AM and 2:00PM ET is matched to a put with identical strike and maturity quoted as closely as possible in time. If the match occurs within the same minute, we use the matched call and put prices as well as the underlying to obtain a candidate short-term asset price for the given maturity; otherwise the match is discarded. The median of the candidate prices is the point estimate of the short-term asset price for the day, and the steepener price is taken as the difference of short-term asset prices.

C. Alternative Strip Price Estimates from OptionMetrics Data

We propose three reasons why strip prices computed from end-of-day OptionMetrics prices should be less noisy than the BBK method of taking medians over intraday MDR prices. First, when $T_1 > 0$, which is the only case where mean logarithmic returns are not biased upwards by taxes, replication requires the synchronization of four separate option prices, two to replicate the synthetic futures on the long side, and two to replicate the synthetic futures on the short side. This synchronization is clearly more reliable when all four option prices are taken from a single end-of-day price matrix than the alternative of attempting to match four intraday time stamps. Second, liquidity is known to be higher at the end of the day than at mid-day, and option spreads are lowest at the end of the day.²⁵ Finally, replication must also use riskless discount rates, and the riskless discount rates used by both BBK and in our study are taken from OptionMetrics. These interest rates are largely based on CME Eurodollar futures timestamped at 3:00PM EST, and therefore match better to the

²⁵See, e.g., Chan, Chung, and Johnson (1995).

end-of-day OptionMetrics prices than to the mid-day (10AM-2PM ET) option quotes used by BBK.²⁶ Whether noise in dividend strip prices can be reduced using end-of-day OptionMetrics data is therefore a promising avenue of empirical investigation.

All strategies we consider use the same rule for choosing T_2 , the short-side of the trade, as BBK. On the long side of the trade we use four alternative instruments. First, following BBK we use option contracts expiring at T_1 as close as possible to 12 months prior to T_2 . Second, we use options with the shortest maturity T_1 that is greater than or equal to 45 days. This rule gives lower leverage, due to the longer maturity claim to dividends of about 15 months. Third, we again set T_1 to the shortest maturity greater than or equal to 45 days, but substitute the actual traded futures price for the synthetic futures price used in the second approach. To synchronize the futures trade timing as closely as possible with the options quotes, we select the last trade price available before 4:15PM ET in the pre-2008:03 data and the last trade price available nearest to 4:00PM ET thereafter.²⁷ The final instrument we consider on the long side of the trade is the closing spot price of the underlying index. Since this closing price is always at 4:00PM ET, there will be a clear mismatch with OptionMetrics data prior to March 2008. We use this primarily to demonstrate the impact of a known timing mismatch.

For the first two strategies, we further consider two different approaches to produce daily estimates of strip prices. First, for each maturity T_1 and T_2 , candidate discounted futures prices are obtained for all put-call pairs with matching strike prices, and we compute the strip price as the difference between the medians of the two maturity groups (method A). Second, we compute all possible candidate strip prices by differencing all combinations of potential long- and short-leg synthetic discounted futures, and take the median across all combinations as the strip price (method B). These differences amount

²⁶BBK consider the idea of using OptionMetrics data to generate strip prices, but only in the context of their short-term asset. See their footnote 6, page 1600. They observe the fifteen-minute mismatch in closing times between the options exchange and the stock market and conclude this method will be problematic. In untabulated results, they report attempting this method, and finding similar strip prices, but much higher volatility. But their investigation is only in the context of the high-yielding and problematic short-term asset. They thereafter abandon the possibility of using OptionMetrics prices and do not consider this for their steepener. For the steepener, or any strategy with $T_1 > 0$, the mismatch between option-market and stock-market closing times is not relevant because stock prices are not used, only option prices.

²⁷Refer to Section IV.A where we previously described the OptionMetrics timestamps.

to the distinction between taking the difference of medians (A) versus the median of differences (B).

An additional benefit of using end-of-day OptionMetrics prices is the wide accessibility of this methodology to other researchers. The intraday MDR data used by BBK is both expensive and not widely used by researchers for other purposes. In addition, their intraday matching technique requires more steps and choices, and is more computationally intensive. Other data used by researchers in this area are proprietary (indicative prices), or suffer from short samples and often low activity (listed futures). In contrast, OptionMetrics data is widely used for other purposes in finance research, and therefore available to many researchers. The methodology using a single end-of-day option matrix for all prices on that day is also easy to implement. Given the importance of the term structure of risk premia for finance research, it is beneficial to have a straightforward methodology for pricing dividend strips using standard data.

D. Empirical Estimates of the Returns to Dividend Strip Strategies

Table 1 shows simple mean returns, standard deviations, first-order autocorrelations, and mean logarithmic returns for the S&P 500 index, the two strategies considered by BBK, and our six new strategies. Panel A provides moments from the daily data in our MDR subsamples, normalized to be in monthly units. Summary statistics for monthly observations during the BBK sample period (02/1996-10/2009) are in Panel B, and for the extended sample period (02/1996-04/2016) in Panel C.

Moments in Panel A show characteristic signs of large measurement error. Mean daily returns exceed 8% monthly, with standard deviations larger than 35% per month and daily autocorrelations less than -0.30. It seems implausible that daily synthetic strip returns convey economically meaningful information or that additional daily data would improve the return estimates.

Summary statistics for monthly returns are also consistent with high levels of noise in strip prices. Panel A documents return standard deviations ranging from 7.8% to 36.4% monthly, which are large relative to the market (4.7%), and monthly autocorre-

lations are strongly negative, ranging from -21% to -37%. Average simple returns are, consequently, much larger than average log returns.

The strips with positive dividend yield, i.e., $1 - \rho > 0$, $T_1 = 0$, have upward biased mean logarithmic returns due to high loadings on heavily taxed dividends (Schulz, 2016). These are also less practical to implement due to the cost of replicating the index (BBK).

The zero-yield strips, $1 - \rho = 0$, $T_1 > 0$, are not exposed to bias due to differential taxation of dividends. Among these strategies, those using only OptionMetrics data have the lowest volatilities and the least negative autocorrelations, consistent with lower noise than the BBK steepener and the strategies using data from the futures market.

The steepeners derived from OptionMetrics data not only have the lowest standard deviations, they also have the lowest continuously compounded returns. In the BBK sample, their log returns are in line with that of the index. In the extended sample, their log returns (around 0.50%) are lower than that of the index (0.66%). The evidence from synthetic dividend strips with the lowest noise and the longest samples therefore does not support a downward-sloping term structure of equity risk premia.

E. Sources of Noise and their Impact on Dividend Strip Returns

We now consider and evaluate the impact of different sources of noise.

E.1. Mistiming in Dividend Strip Components

Asynchronicity in prices is a well-known source of error when constructing synthetic payoffs that require cross-market positions. All of the strip strategies that we consider combine information from options and bonds markets, and sometimes additionally require pricing information from the cash or futures markets.

The first, and most direct source of noise is an explicit non-synchronicity in the timing of quotes/prices underlying the synthetic strategy. The strip that combines the 4:00PM underlying index price with option quotes from 4:15PM illustrates this effect. In particular, mistiming will result in a pricing error today that is negatively correlated

with the unobserved return of the underlying between 4:00-4:15PM. This pricing error will be reversed tomorrow, and therefore we expect a positive coefficient when regressing the next month's strip return on the 4:00-4:15PM return in the underlying today.

In untabulated results, we verify this by approximating the index return between 4:00 and 4:15PM using the corresponding return of the nearest-to-maturity futures. In univariate regressions using our entire sample, this noise proxy positively predicts next month's strip return with a t -statistic of 3.5 and an adjusted R^2 of 6.4%. The coefficient remains significant after controlling for the lagged strip return, and the regression R^2 increases to 15%. Using the pre-2008:03 data where OptionMetrics quotes are from 4:15PM, these results become stronger: the t -statistic on noise rises to 3.85 with an adjusted $R^2 = 8.8\%$ and in the regression that includes lagged strip return adjusted R^2 rises to 16.73%. In the post-2008:03 sample the regressions become uninformative (e.g., in the regression with lags adjusted $R^2 = 1\%$.) Due to the obvious timing problem, we drop this strategy from further analysis.

A less obvious, but equally consequential example of timing mismatch shows up in the BBK strip when they match mid-day option and underlying prices with the 3:00PM ET Eurodollar futures settlement prices. As a result, the interest rate used is on average timestamped three hours after the options prices.

While BBK argue that taking the median of many individual quotes should eliminate noise, this is only true if the noise in individual quotes is iid. However, the mismatch in timing implies that BBK are discounting all their futures prices at the wrong rate, leading to correlated errors that are not fixed by taking the median. We therefore expect the change in risk-free rate between the timestamp of the options contract and 3:00PM ET to predict returns. If this interest rate change is positive, BBK use too large a discount rate. This discount rate affects the short side of the trade (long maturity) more than long side (short maturity). A positive rate change therefore predicts too low a price on the short side of the trade, and should negatively predict next months return.

We verify this in Table 2. While we do not have access to the exact timestamp of the selected BBK option or intraday data for the relevant risk-free rate, we proxy for the rate change since mid-day by the change in the interest rate applicable to the long horizon (short leg) option from the previous end of day. The results are striking: The change in the risk-free rate between the second-last day to the last day of the month explains over 15% of the variation in next month’s returns of the BBK strategies. The intercepts suggest that when interest rates do not move on the last day of the month, the mean returns are expected to be more than one third lower than the unconditional values in Table 1 (0.76% vs. 1.16% and 0.65% vs. 1.12%).

Changes in interest rates also relate negatively to the future returns of our Option-Metrics strategies. However, the economic effect is much weaker (R^2 all less than 5.1%) because the timing mismatch is much smaller than for the BBK strategies.

The coefficients on the interest rate change remain significant when controlling for return autocorrelations. As expected, the negative autocorrelations also decline strongly after controlling for the interest-rate-timing mismatch. However, autocorrelations remain negative and significant, suggesting other channels of noise are important.

E.2. A Measure of Noise in Synthetic Strip Prices and Reversal Tests

Our previous analysis is based on the assumption that the bid-ask midpoints are, on average for a given day, equal to the fundamental option prices. We now use the dispersion of candidate strip prices to investigate the impact of noise in the median price estimate on synthetic strip return moments.

Let

$$\eta_t = \ln \left[\frac{\text{std}(V_{t,T_1,T_2}^i)}{V_{t,T_1,T_2}} \right], \quad (26)$$

represent the relative dispersion in strip prices, where V_{t,T_1,T_2}^i are all candidate strip prices on day t , and V_{t,T_1,T_2} is their median.²⁸

²⁸In the cases where strip prices are computed as difference between two futures price medians (method A), our noise measure is the average of the noise measures for each of the two legs.

Because we estimate only the quantity of noise, we do not have directional predictions. Instead, we test if reversals are stronger when prices are noisier. In particular, we run the regression

$$R_t = \beta_0 + \beta_1 R_{t-1} + \beta_2 \eta_{t-1} + \beta_3 (R_{t-1} \times \eta_{t-1}) + \varepsilon_t, \quad (27)$$

where the noise proxy is standardized to have zero mean, and test whether β_3 is negative.

The results are in Table 3. Panel A again shows regressions of daily returns on the three subsamples. We first note that the coefficient on the interaction term is negative and significant, indicating that reversal is stronger on days with more noise. Second, since η_t is demeaned, we can also interpret the coefficient on lagged returns. On days with an average amount of noise, the conditional autocorrelation is about one third lower than unconditional (-0.24 vs. -0.34, -0.20 vs. -0.30). Lastly, the Blume and Stambaugh (1983) return bias manifests itself in a significantly positive coefficient on η_t , confirming that noise biases mean simple returns upwards.

The findings for the monthly return series in Panels B and C are similar. Noise significantly increases reversal. Moreover, for the OptionMetrics strategies the coefficients on lagged returns drop dramatically and become statistically insignificant after controlling for noise.

V. Implications for the Term Structure of Equity Risk Premia

We summarize our main findings regarding the term structure of equity risk premia and the relevance for the asset pricing literature.

A. Noise in synthetic dividend strip prices is large, and not eliminated by taking price medians.

One might wonder why taking medians across many candidate strip price estimates, as in BBK, does not completely eliminate noise. Figure 4 shows implied bid-ask spreads of synthetic dividend strips. Specifically, we approximate the spreads of the BBK short-term asset, which is long the index and short a synthetic futures. The synthetic futures

price is obtained from the prices of puts and calls by put call parity. We ignore the index spread since it is small relative to the other legs of the trade. To approximate the dollar spread on a synthetic futures, we simply add the spreads of a call and a put with the same strike. Panel A shows the absolute values of the implied spreads on these strips as a function of maturity. Throughout most of the sample, spreads cluster at even dollar amounts of \$4, \$6, or even \$8 at long maturities. For comparison, one-year dividend strips range in price from \$7 to \$32, so the spread and the midpoint have a similar order of magnitude. Panel B shows the bid-ask spreads as a percentage of strip value. For the BBK short-term asset, relative bid-ask spreads range from 10-60% of the estimated strip value.²⁹

As a consequence of the large spreads shown in Figure 4, and the systematic sources of noise documented earlier, the range of candidate strip prices are very large. Figure 5 shows all candidate short-term asset prices in March 2001. Options with approximately 18 months to maturity (blue) are used to construct the short leg of both the BBK short-term asset and the steepener, options with 6 months to maturity (red) are required for the long leg of the steepeners. For any given day, the range of candidate prices can vary from prices less than \$0 to more than \$20. Taking the median over many of these candidate prices does reduce noise, but does not eliminate noise.

B. Predictability in dividend strip returns is evidence of noise, not time-varying risk premia.

Table 1 shows very strong negative autocorrelations in dividend strip returns at monthly frequencies. BBK interpret these price reversals as evidence of time-varying risk premia. (See their Sections III.D. and III.E.)

Maheswaran and Sims (1993) consider the possibility of price reversals in arbitrage-free markets. They show that “*the martingale equivalence condition delivered by a no-arbitrage assumption in complete asset markets has implications for fine-time-unit asset price behavior that can be rejected with finite spans of data.*” In particular, infi-

²⁹It is interesting to note that the lowest percentage spreads occur in the earliest subsample of their data, not later as would occur if pricing were becoming more efficient.

nite variation in a finite amount of time, i.e., too many reversals in too short a time, violates the no-arbitrage condition. In other words, while reversals themselves are not inconsistent with absence of arbitrage, as one goes to higher frequencies, at some point risk premia must exhibit positive or zero persistence, and scaled return variances must remain bounded.

Figure 6 shows variance ratio as a function of the return-measurement interval. In these plots, all variance ratios increase as the measurement interval became smaller, to the shortest interval of one day. This evidence is exactly the test prescribed by Maheswaran and Sims (1993), and is inconsistent with time-varying risk premia in arbitrage-free markets. The test implies the presence of arbitrage opportunities, which could be tradeable, or could be due to noise.³⁰

C. Simple mean returns can be very misleading in the presence of noise.
We can manufacture strategies to show either an upward- or downward-sloping term structure in simple mean returns by manipulating levered noise.

In Section III, we showed that the mean returns of standard pairs trades is U-shaped in leverage, increasing for either very large positive, or very large negative values of leverage. We can similarly manipulate the amount of noise in dividend strip trading strategies.

Figure 7 shows mean simple (red) and log (black) returns of portfolios that are *long* $\mathcal{L} + 1$ units of dividend strips exposed to dividends in approximately the range 18 to 24 months, and *short* $\mathcal{L}$ units of dividend strips exposed to dividends in the range from 6 to 12 months. The figure shows that while log returns are largely unaffected by leverage, mean simple returns increase with both negative and positive leverage.

It is important to remember that when leverage $\mathcal{L}$ is positive, we have positive exposure to the longer maturity contract, and when $\mathcal{L}$ is negative, we have positive

³⁰One might interpret Figure 6 to imply a need for models that admit arbitrage opportunities, perhaps by contemplating heterogeneous agents with different trading costs and/or information sets. The high-frequency return moments of the data would likely be more informative about the unique mechanisms of the microstructure aspects of such a model, as opposed to lower-frequency moments that would be more informative about sources of risk from traditional consumption-based asset pricing models.

exposure to the shorter maturity contract. If simple returns were informative about the slope of the term structure, we would see a monotonic relationship between leverage $\mathcal{L}$ and simple returns. We do not see a monotonic relationship between leverage $\mathcal{L}$ and mean simple returns, but rather it is U-shaped. We conclude that in the presence of large amounts of noise, simple mean returns are uninformative about the slope of the term structure.

D. Using improved empirical methodologies and longer samples of data that are more widely available to other researchers, the cumulative returns to synthetic dividend strips are *lower* than returns on the market. There is *no support* for an unconditional downward-sloping term structure.

Our paper emphasizes the importance of noise to the return moments of synthetic replication strategies and develops methods to reduce the impact of noise. We recommend that inference focus on:

- *mean logarithmic returns*, which are less affected by noise than mean simple returns;
- trading strategies that *do not collect dividends*, and hence are not confounded by tax differences (Schulz, 2016), and the cost and difficulty of instantaneously replicating the underlying basket of securities in the index, as well as reinvesting dividends;
- trading strategies that are based on *end-of-day option prices*, which are less contaminated by a timing mismatch with CBOE interest rates than the BBK methodology, have lower return variance and less negative autocorrelation than the BBK steepener, and are easier for other researchers to replicate using widely available OptionMetrics data.

To further demonstrate the impact of these methodological and data improvements, Figure 8 displays the full time series of cumulative returns for the S&P 500 (black line) and several synthetic dividend strategies. The red lines show the hypothetical

cumulative returns one obtains from compounding the mean simple monthly returns reported in BBK. For example, the mean simple monthly returns are 1.16% and 1.12% for the short-term asset (solid line) and steepener (dashed line) in comparison with the mean simple S&P 500 index return of 0.58%. These mean simple returns (see also our Table 1) falsely give the impression of a downward-sloping term structure, and *all* of the tables and figures in BBK are based on comparisons and statistics computed from simple monthly returns. The explosive growth of the red lines is unrealistic because of the upward bias in the mean simple returns of the noisy estimated dividend strip returns. As is well-known, (e.g., Blume and Stambaugh, 1983), it is inappropriate to compare the mean simple returns of a noisy series with the mean simple returns of a less noisy series.

The orange lines show the actual cumulative returns of the two BBK series. As anticipated these are well-below the red lines derived from compounding mean simple returns. However, the cumulative return series are still noisy, and potentially biased. The higher of the two orange lines (dashed), representing the BBK short-term asset, is particularly problematic. This strategy has the market index on the long side and a high average dividend yield exceeding 5% per month. As a consequence of the high dividend yield, even the average logarithmic return of this strategy is upward biased, because of differential taxation of dividends versus capital gains (Schulz, 2016). The short-term asset strategy faces additional difficulties in implementation because of the need to instantaneously purchase the entire basket of securities in the underlying index and reinvest dividends (BBK). The lower of the two orange lines (solid), representing the BBK steepener, does not face these difficulties as it does not collect dividends. The steepener strategy does still have noisy returns, as we demonstrate in Section IV, due to the timing mismatch between interest rate measurement and option price measurement, the difficulty of matching in time the four distinct options underlying the strategy, and the high spreads on those options.

The four green lines demonstrate the cumulative returns to the dividend strip strate-

gies that implement all of our recommendations. These strategies use end-of-day prices from OptionMetrics, which reduces the timing mismatch with measured interest rates and the timing mismatch between options, and readily allows extension of the sample period. Further, in comparison with the BBK steepener, all four of these series have lower standard deviations and autocorrelations closer to zero, both of which are indicative of less noise. (See Table 1.) The cumulative returns are lower than the BBK steepener, and over the full sample, all have lower cumulative returns than the S&P 500 index.

E. Our analysis helps to clarify the validity of the existing empirical evidence on the term structure of equity risk premia.

BBK, Schulz (2016), and our study are based on synthetic dividend strip prices. BBK and Schulz use intraday MDR option prices, while our paper demonstrates the advantages of using end-of-day OptionMetrics data. Binsbergen, Hueskes, Koijen, and Vrugt (2013), Binsbergen and Koijen (2017), and Bansal, Miller, Song, and Yaron (2018) work with indicative, and in some cases traded, dividend futures prices. The different data sources are complementary.

One clear advantage of the OptionMetrics data is that it allows us to create the longest internally consistent time-series of dividend strip returns of any study. Our data sample covers January 1996 - April 2016. Binsbergen, Hueskes, Koijen, and Vrugt (2013) use data from October 2002 - March 2011. Binsbergen and Koijen (2017) update this data to October 2002 - July 2014. Bansal, Miller, Song, and Yaron (2018) use data from December 2004 - February 2017. Given the short time series, extending coverage where possible is important, and our study provides the longest sample available.

A second advantage of the OptionMetrics data is its wide availability. The primary data sources for dividend futures prices in extant studies are proprietary dealer databases.³¹ For the S&P 500, proprietary databases are the only source of futures

³¹Binsbergen, Hueskes, Koijen, and Vrugt (2013) and Binsbergen and Koijen (2017) use data from BNP Paribas and Goldman Sachs, while Bansal, Miller, Song, and Yaron (2018) do not specify the financial institution that provides their data.

prices (other than synthetic pricing) for the majority of the sample period. Dividend futures first became listed in the United States in October 2015, on the Chicago Mercantile Exchange. Prior to that date, over-the-counter (OTC) swaps were available.³² S&P 500 dividend futures prices can be downloaded from Bloomberg from the time of their initial listing in October 2015.³³ While proprietary databases can be useful, when researchers have questions outside the scope of the tables and figures in published work, access to widely available underlying data is an important advantage.

Third, it is useful to compare results from synthetic dividend strips to those from traded dividend futures since both relate to the same fundamentals, but are affected by different sources of noise. We have carefully documented important sources of errors in synthetic dividend strip prices related to our theory of levered noise, and the resulting impacts on returns. While our theory of levered noise does not apply to directly observed trade prices, quotes, or indicative prices, this does not mean that those prices or quotes are free of noise. A large prior literature has studied the effects of noise in directly observed prices and quotes (e.g., Blume and Stambaugh, 1983).

Recent studies document important sources of noise in traded dividend futures and swaps, and in the indicative prices maintained by participants in those markets. Mixon and Onur (2017), written by two economists from the Commodity Futures and Trading Commission, uses comprehensive regulatory data on actual trades in dividend futures and swaps. They focus on an 18-month sample beginning in 2015, and estimate that their data covers over 70% of all trades with an S&P 500 dividend underlying the contract. They find that of the 375 days of coverage in their sample, only 34 (less than 10%) have any trade, in any maturity of contract, between a “dealer” and an “end-user.”³⁴ Further, only 156 days (about 40%) have any trades, in any maturity of contract, between dealers. They conclude, “*markets are not particularly active beyond*

³²See Mixon and Onur (2017) for further discussion. The Chicago Board Options Exchange listed S&P 500 dividend options in March 2010 but these have not been actively traded.

³³Bansal, Miller, Song, and Yaron (2018) state that they obtain S&P 500 dividend futures prices from Bloomberg beginning in January 2010. These would have to be over-the-counter prices and the ticker is not readily apparent from a Bloomberg query.

³⁴They categorize institutions into dealers and end-users. End-users for example include pension funds, hedge funds, and insurance companies.

the first few years' worth of dividends, and [...] OTC transactions with customers occur sporadically. This is particularly a problem with S&P 500 dividends, where no futures market existed until late 2015. Analyses using indicative OTC dividend swap prices should be clear about these features of the market."³⁵ Bansal, Miller, Song, and Yaron (2018) express related concerns about low liquidity and large spreads in their data. It is therefore critical to have evidence from other sources, especially from data that is widely available to other researchers, such as the synthetic price and return series that we develop.

Finally, when we compare the evidence from our study with the evidence from indicative futures prices, a consistent story emerges: There is no evidence to support a downward-sloping term structure of equity risk premia, and the standard errors around any inference are very large. Early impressions in the literature were heavily influenced by the highly biased mean simple returns at the heart of all of the tables and figures in BBK, and the other seemingly anomalous moments reported in that study including high volatility and high predictability of dividend strip returns. Our theory of levered noise explains the biases in those results. Since the initial BBK study, evidence primarily from financial institution indicative prices, supplemented by limited data from exchange-traded futures, provides a different view. Binsbergen, Hueskes, Koijen, and Vrugt (2013) and Binsbergen and Koijen (2017) argue that their data supports an unconditional downward-sloping term structure, but the actual evidence in those papers is very mixed across maturities and markets, and much weaker than the biased evidence in the initial BBK study. Bansal, Miller, Song, and Yaron (2018) use similar data to come to the opposite conclusion of an upward-sloping unconditional term structure. We provide complementary evidence. While the standard errors on any inference are large, our best full-sample point estimates of the returns to dividend strips in the U.S. are lower than those of the market over the same period, in the range of 47-58 basis points per month for logarithmic dividend strip returns, versus 66 basis point per month for

³⁵Mixon and Onur (2017) explain that OTC transactions are generally swaps and exchange-traded contracts are generally futures.

the logarithmic index return.

VI. Conclusion

We show theoretically and empirically that small pricing frictions in primary markets become greatly magnified in importance when one uses highly levered long-short positions to replicate and synthetically price financial claims. Such amplification implies large biases in the moments of synthetic return series. We apply these ideas to dividend strips, and show that the magnification of pricing errors implies upward-biased short-horizon returns and negative autocorrelation, consistent with empirical evidence. Using improved empirical methodologies that account for noise, and longer samples of more standard data, we find that cumulative returns to synthetic dividend strip strategies are *lower than* returns on the market index. There is no evidence of an unconditional downward-sloping term structure.

The concept of levered noise is more generally important as another form of limits to arbitrage. Many standard arbitrage trades, such as the replication of calls or other options, and a variety of calendar- and other types of spreads, involve high leverage. Seemingly negligible frictions of a few basis points in primary markets can imply substantial pricing errors of several percentage points or more in synthetically priced derivative assets, and can therefore limit the effectiveness of highly levered arbitrage trading strategies.

Appendix

Derivation of Equation (5)

Analysis of the impact on returns of noise in prices is greatly facilitated by transforming returns to their logarithmic approximations. Denote the expected log per-period dividend yield by $\overline{d-v}$ and further define $\rho = 1/(1 + \exp(\overline{d-v}))$ and $k = -(1 - \rho) \log(1/\rho - 1) - \log(\rho)$. Then

$$\begin{aligned}
r_{t,1} &= \log(R_{t,1}) \\
&= \log\left(\frac{V_{t+1} + D_{t+1}}{V_t}\right) \\
&= \log\left[\left(\frac{V_{t+1}}{V_t}\right) \left(1 + \frac{D_{t+1}}{V_{t+1}}\right)\right] \\
&= v_{t+1} - v_t + \log\left(1 + e^{\overline{d-v}} e^{d_{t+1} - v_{t+1} - \overline{d-v}}\right) \\
&= k + \rho(v_{t+1} - v_t) + (1 - \rho)(d_{t+1} - v_t)
\end{aligned} \tag{28}$$

where the last line utilizes the first-order approximation

$$f(x) = \log(1 + e^a e^x) \approx \log(1 + e^a) + \frac{e^a}{1 + e^a} x. \tag{29}$$

Derivation of Equations (10) and (11)

We begin by log-linearizing the noise-free price. Assuming $V_t > 0$ and $\mathcal{L} > -1$, and denoting $l_t = \log(L_t)$ and $s_t = \log(S_t)$, we may compute

$$\begin{aligned}
v_t &= \log(V_t) \\
&= \log[L_t + \mathcal{L}(L_t - S_t)] \\
&= \log\left[L_t \left(1 + \mathcal{L} \left(1 - \frac{S_t}{L_t}\right)\right)\right] \\
&= \log\left[(1 + \mathcal{L})L_t \left(1 - \frac{\mathcal{L}}{1 + \mathcal{L}} e^{s_t - l_t}\right)\right] \\
&= \log(1 + \mathcal{L}) + l_t + \log\left(1 - \frac{\mathcal{L}}{1 + \mathcal{L}} e^{s_t - l_t}\right) \\
&= l_t + \mathcal{L}(l_t - s_t),
\end{aligned} \tag{30}$$

where the last line in the derivation uses the approximation in equation (29). Equation (10) follows from equation (30) using the substitutions $l_t^o = l_t + \varepsilon_{L,t}$ and $s_t^o = s_t + \varepsilon_{S,t}$. Values of $\mathcal{L} < -1$ lead to the same conclusion since the labelling of L_t and S_t is arbitrary. The equation holds trivially when $\mathcal{L} = -1$.

Equation (11) follows from the same logic leading to equation (30) with the appropriate notational changes for dividends.

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Table 1. RETURN MOMENTS OF DIVIDEND STRIPS

	Securities		Approximate maturity, months		Moments			
	Long	Short	Long	Short	Mean	Stddev	AR(1)	Log mean
A. Daily subsamples								
BBK short-term asset	index	options	0	18	8.55	37.0	-0.34	1.90
BBK steepener	options	options	6	18	8.63	43.4	-0.30	-0.66
B. Monthly BBK sample								
S&P 500 index	index				0.58	4.68	0.13	0.46
<i>Positive Dividend Yield: $(1 - \rho) > 0$, $T_1 = 0$</i>								
BBK short-term asset	index	options	0	18	1.16	7.80	-0.27	0.85
OptionMetrics-index strip	index	options	0	18	1.52	12.1	-0.37	0.81
<i>Zero Dividend Yield: $(1 - \rho) = 0$, $T_1 > 0$</i>								
BBK steepener	options	options	6	18	1.12	9.65	-0.31	0.65
OptionMetrics strip 1A	options	options	6	18	0.82	8.79	-0.21	0.42
OptionMetrics strip 1B	options	options	6	18	0.85	8.77	-0.22	0.46
OptionMetrics strip 2A	options	options	3	18	0.88	7.85	-0.23	0.56
OptionMetrics strip 2B	options	options	3	18	0.84	7.87	-0.24	0.52
OptionMetrics-futures strip	futures	options	3	18	3.10	36.4	-0.23	0.18
C. Monthly extended sample								
S&P 500 index	index				0.76	4.41	0.07	0.66
<i>Positive Dividend Yield: $(1 - \rho) > 0$, $T_1 = 0$</i>								
OptionMetrics-index strip	index	options	0	18	1.43	10.3	-0.37	0.91
<i>Zero Dividend Yield: $(1 - \rho) = 0$, $T_1 > 0$</i>								
OptionMetrics strip 1A	options	options	6	18	0.76	7.45	-0.21	0.47
OptionMetrics strip 1B	options	options	6	18	0.78	7.42	-0.21	0.50
OptionMetrics strip 2A	options	options	3	18	0.81	6.63	-0.22	0.58
OptionMetrics strip 2B	options	options	3	18	0.78	6.65	-0.23	0.55
OptionMetrics-futures strip	futures	options	3	18	2.43	31.3	-0.23	0.24

Notes: This table reports the securities underlying different dividend strip strategies and their approximate maturity as well as simple means, standard deviations, autocorrelations, and log means of returns. Moments for the S&P 500 index are also reported. Panel A is based on daily returns implemented using MDR data on the subsamples 01/1996-07/1996, 10/2000-10-2001, and 01/2009-07/2009. Panel B uses monthly return frequencies in the BBK sample (02/1996-10/2009) and Panel C extends the sample to 04/2016. The short leg of all strips uses end-of-day OptionMetrics prices for maturities with T_2 , in years, satisfying $1.3 \leq T_2 \leq 1.9$. Strips denoted “1” and “2” differ by maturity of the long synthetic futures position: 1A and 1B strips use options with maturity nearest to 12 months before the long-dated option maturities; 2A and 2B strips use options with the shortest maturity $T_1 \geq 45$ days. Strip prices “1A” and “2A” are constructed by differencing the median prices of the long and short synthetic futures positions, while strip prices “1B” and “2B” equal medians of all candidate strip prices that can be computed from end-of-day, strike- and maturity-matched OptionMetrics prices. The dividend yield of the strip strategies is denoted $(1 - \rho)$ (see Equation (5)).

Table 2. EVIDENCE OF NOISE FROM RISK-FREE RATE MISTIMING

	Intercept	$\Delta R_{\tau-1:\tau}^F$	Adj R ²	Intercept	R_{t-1}	$\Delta R_{\tau-1:\tau}^F$	Adj R ²
A. Daily subsamples							
BBK short-term asset	0.21 [0.69]	-0.34 [-7.38]	7.53	0.38 [1.27]	-0.27 [-6.57]	-0.20 [-4.11]	13.23
BBK steepener	0.14 [0.41]	-0.45 [-8.41]	9.60	0.29 [0.83]	-0.23 [-5.72]	-0.33 [-5.70]	13.89
B. Monthly BBK sample							
BBK short-term asset	0.76 [1.36]	-0.48 [-5.70]	16.11	1.03 [1.84]	-0.21 [-2.96]	-0.43 [-5.21]	19.74
BBK steepener	0.65 [0.93]	-0.57 [-5.49]	15.06	0.96 [1.40]	-0.24 [-3.41]	-0.49 [-4.78]	20.06
OptionMetrics-futures strip	3.04 [1.07]	-0.64 [-0.81]	-0.21	3.80 [1.36]	-0.23 [-3.04]	-0.69 [-0.89]	4.65
OptionMetrics strip 1A	0.59 [0.87]	-0.28 [-2.79]	3.98	0.76 [1.13]	-0.19 [-2.53]	-0.26 [-2.58]	7.02
OptionMetrics strip 1B	0.62 [0.92]	-0.28 [-2.77]	3.92	0.82 [1.22]	-0.20 [-2.69]	-0.26 [-2.59]	7.46
OptionMetrics strip 2A	0.65 [1.08]	-0.28 [-3.08]	4.92	0.85 [1.43]	-0.21 [-2.82]	-0.25 [-2.85]	8.78
OptionMetrics strip 2B	0.61 [1.01]	-0.28 [-3.13]	5.09	0.81 [1.36]	-0.22 [-2.96]	-0.26 [-2.89]	9.38
C. Monthly extended sample							
OptionMetrics-futures strip	2.38 [1.14]	-0.58 [-0.87]	-0.11	2.96 [1.45]	-0.23 [-3.54]	-0.63 [-0.96]	4.80
OptionMetrics strip 1A	0.52 [1.09]	-0.27 [-3.30]	3.93	0.67 [1.43]	-0.19 [-3.04]	-0.25 [-3.03]	7.05
OptionMetrics strip 1B	0.55 [1.16]	-0.27 [-3.27]	3.86	0.72 [1.53]	-0.20 [-3.16]	-0.25 [-3.05]	7.32
OptionMetrics strip 2A	0.58 [1.37]	-0.26 [-3.56]	4.61	0.76 [1.82]	-0.20 [-3.29]	-0.24 [-3.30]	8.32
OptionMetrics strip 2B	0.55 [1.31]	-0.27 [-3.62]	4.75	0.73 [1.76]	-0.21 [-3.46]	-0.24 [-3.34]	8.85

Notes: This table reports results of regressions of returns of dividend strip strategies on their lag and the change in interest rates on the last day prior to the observation, $\Delta R_{\tau-1:\tau}^F$. Panel A is based on daily returns implemented using MDR data on the subsamples 01/1996-07/1996, 10/2000-10-2001, and 01/2009-07/1009. Panels B and C use monthly return frequencies in the BBK sample (02/1996-10/2009) and the sample extended to 04/2016.

Table 3. VARIATION IN IMPUTED PRICES AS A PROXY FOR NOISE

	Intercept	R_{t-1}	Adj R ²	Intercept	R_{t-1}	η_{t-1}	$\eta_{t-1} \times R_{t-1}$	Adj R ²
A. Daily subsamples								
BBK short-term asset	0.396 [1.23]	-0.34 [-8.44]	11.36	0.36 [1.12]	-0.24 [-4.32]	0.02 [3.23]	-0.14 [-2.23]	14.06
BBK steepener	0.401 [1.05]	-0.30 [-7.54]	9.29	0.34 [0.89]	-0.20 [-3.24]	0.02 [3.39]	-0.15 [-2.06]	12.06
B. Monthly BBK sample								
OptionMetrics-futures strip	3.83 [1.37]	-0.23 [-3.03]	4.78	0.57 [0.22]	-0.57 [-6.19]	0.01 [0.31]	-0.47 [-5.91]	22.00
OptionMetrics strip 1A	0.98 [1.44]	-0.21 [-2.72]	3.78	0.53 [0.80]	-0.07 [-0.88]	0.01 [0.93]	-0.23 [-3.02]	8.77
OptionMetrics strip 1B	1.04 [1.53]	-0.22 [-2.85]	4.20	0.61 [0.91]	-0.09 [-1.09]	0.01 [0.90]	-0.22 [-2.77]	8.40
OptionMetrics strip 2A	1.07 [1.77]	-0.23 [-3.02]	4.76	0.64 [1.08]	-0.08 [-0.90]	0.01 [1.56]	-0.14 [-2.25]	9.04
OptionMetrics strip 2B	1.03 [1.71]	-0.24 [-3.17]	5.26	0.58 [0.98]	-0.08 [-0.90]	0.01 [1.60]	-0.16 [-2.56]	9.91
C. Monthly extended sample								
OptionMetrics-futures strip	2.99 [1.46]	-0.23 [-3.52]	4.84	0.58 [0.31]	-0.56 [-7.18]	0.01 [0.57]	-0.47 [-6.86]	21.92
OptionMetrics strip 1A	0.90 [1.90]	-0.21 [-3.27]	3.87	0.55 [1.19]	-0.08 [-1.09]	0.01 [1.11]	-0.23 [-3.65]	9.06
OptionMetrics strip 1B	0.94 [2.01]	-0.21 [-3.37]	4.11	0.60 [1.30]	-0.09 [-1.23]	0.01 [1.08]	-0.22 [-3.38]	8.61
OptionMetrics strip 2A	0.98 [2.33]	-0.22 [-3.53]	4.55	0.63 [1.54]	-0.08 [-1.05]	0.01 [1.95]	-0.14 [-2.77]	9.26
OptionMetrics strip 2B	0.95 [2.27]	-0.23 [-3.70]	5.00	0.59 [1.44]	-0.08 [-1.04]	0.01 [1.94]	-0.17 [-3.15]	10.02

Notes: This table reports results of regressions of the return of dividend strip strategies (R_t) onto its lag, a proxy for the noise in the strip prices (η_t), and an interaction term:

$$R_t = \beta_0 + \beta_1 R_{t-1} + \beta_2 \eta_{t-1} + \beta_3 (R_{t-1} \times \eta_{t-1}) + \varepsilon_t,$$

The noise proxy is the standard deviation of prices of dividend strips candidates relative to their median:

$$\eta_t = \ln [\text{std} (V_{t,T_1,T_2}^i) / V_{t,T_1,T_2}] ,$$

Candidate strip prices are based on put-call parity using options with different strike prices (all Panels) and sampled at different times of the day (Panel A only). In the cases where strips are computed as difference between two short term assets, noise is averaged over the two legs. Panel A is based on daily returns implemented using MDR data on the subsamples 01/1996-07/1996, 10/2000-10-2001, and 01/2009-07/1009. Panels B and C use monthly return frequencies in the BBK sample (02/1996-10/2009) and the sample extended to 04/2016.

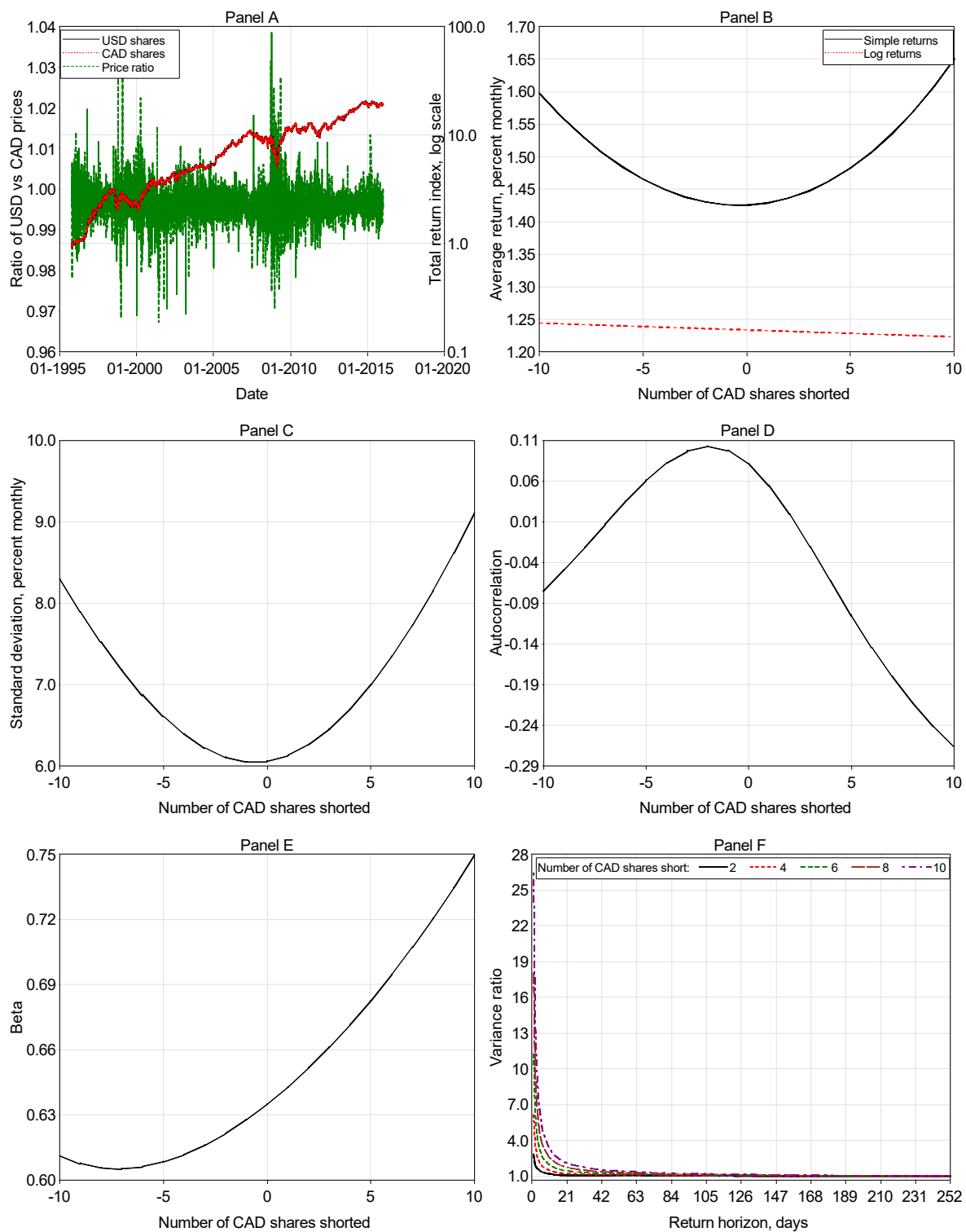


Figure 1. The Effects of Leverage on Returns: Case I, Cross Listing. This figure plots the ratio of prices of Royal Bank shares traded in NY and TSX as well as their total return in Panel A. Panels B-E show how the following return statistics relate to the number of TSX shares shorted $\mathcal{L}$, holding the net investment constant at one share: average simple and log returns (Panel B), return standard deviations (Panel C), autocorrelations (Panel D), and betas (Panel E). Panel F shows variance ratio plots for varying leverage levels. The sample period is October 1995 to December 2015.

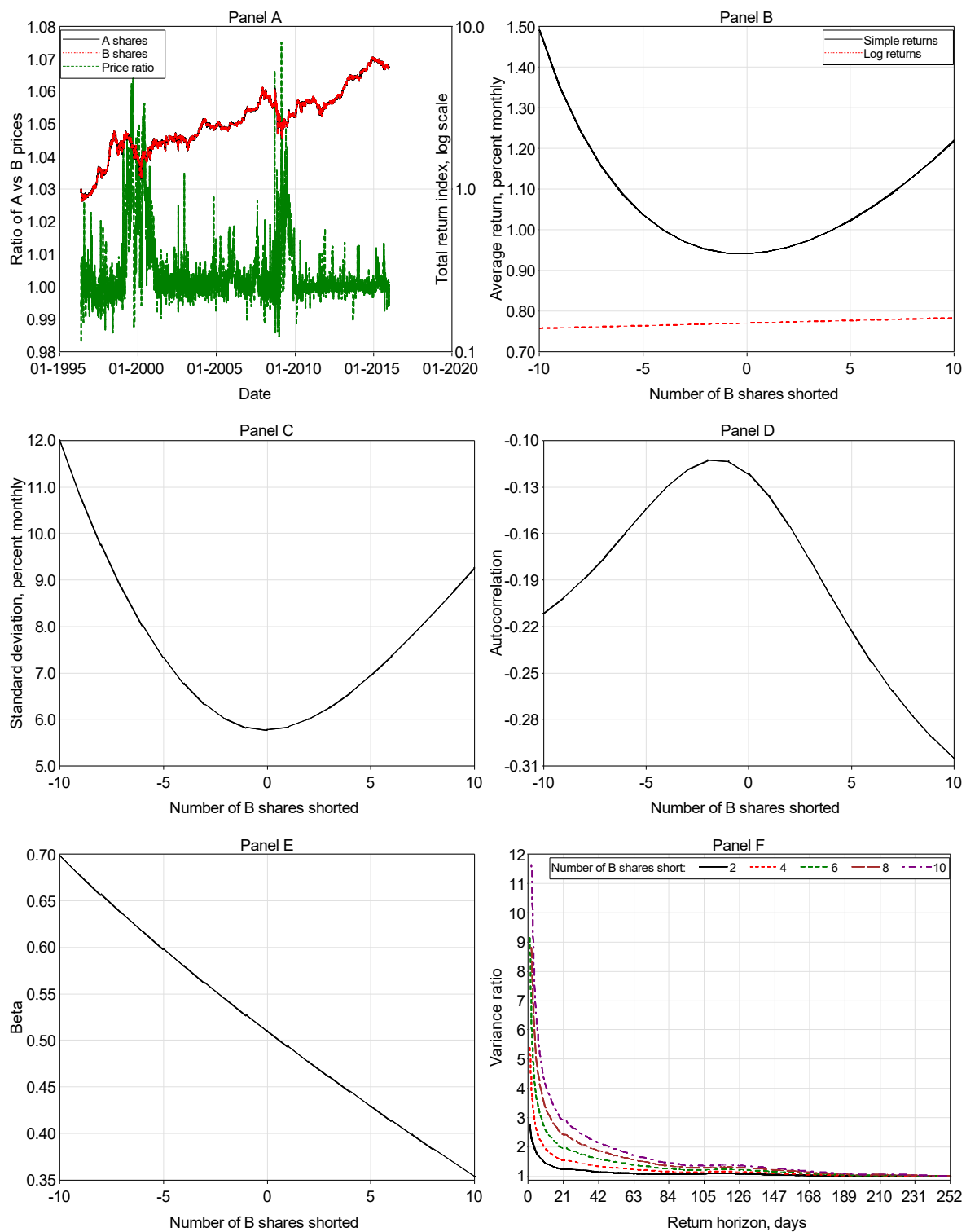


Figure 2. The Effects of Leverage on Returns: Case II, Berkshire Hathaway Dual Class Shares. This figure plots the ratio of prices of Berkshire Hathaway A and B shares as well as their total return in Panel A. Panels B-E show how the following return statistics relate to the number of class B shares shorted $\mathcal{L}$, holding the net investment constant at one share: average simple and log returns (Panel B), return standard deviations (Panel C), autocorrelations (Panel D), and betas (Panel E). Panel F shows variance ratio plots for varying leverage levels. The sample period is March 1996 to December 2015.

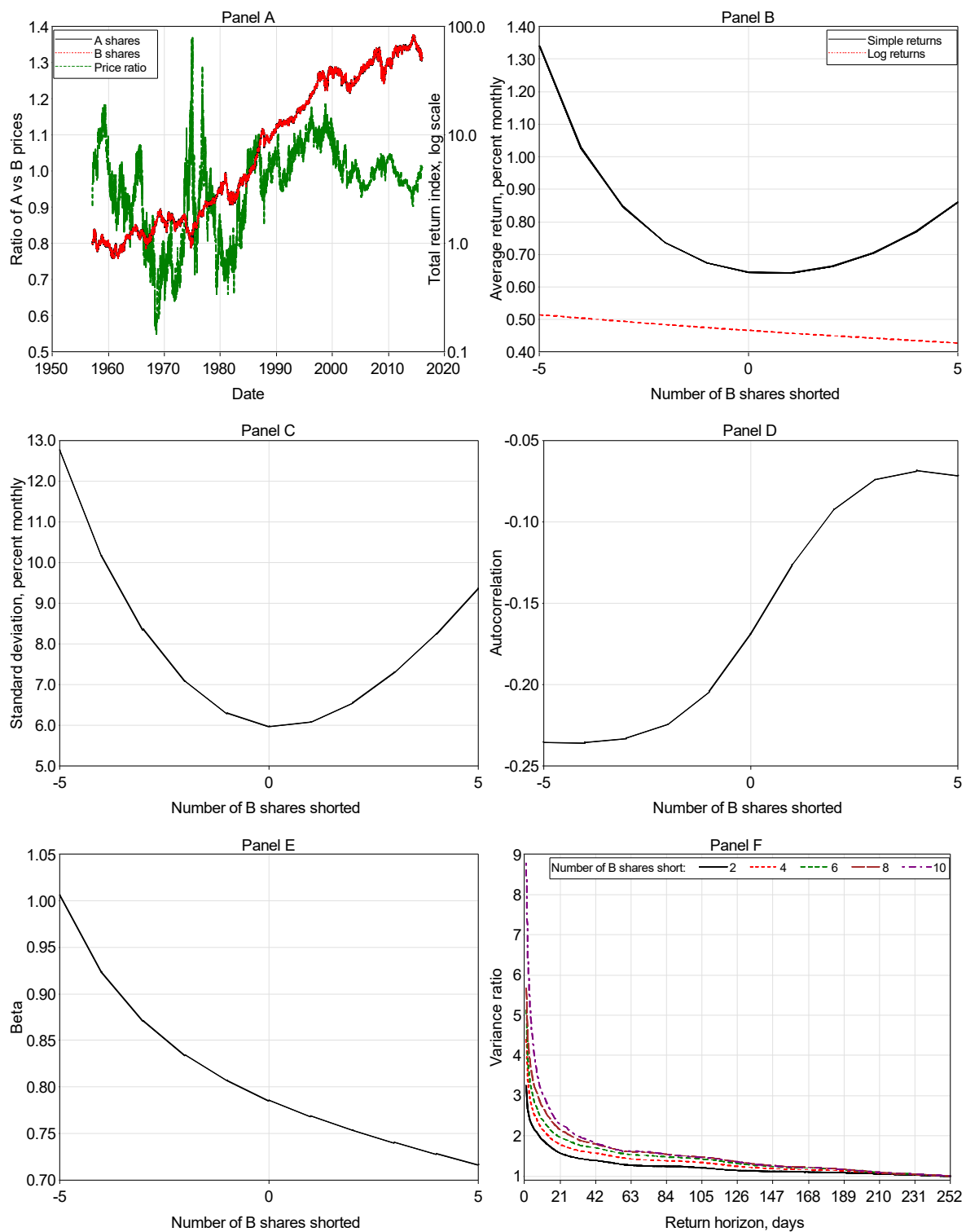


Figure 3. The Effects of Leverage on Returns: Case III, Royal Dutch Shell. This figure plots the ratio of prices of Royal Dutch Shell A and B shares traded in NY as well as their total return in Panel A. Panels B-E show how the following return statistics relate to the number of class B shares shorted $\mathcal{L}$, holding the net investment constant at one share: average simple and log returns (Panel B), return standard deviations (Panel C), autocorrelations (Panel D), and betas (Panel E). Panel F shows variance ratio plots for varying leverage levels. The sample period is March 1957 to December 2015.

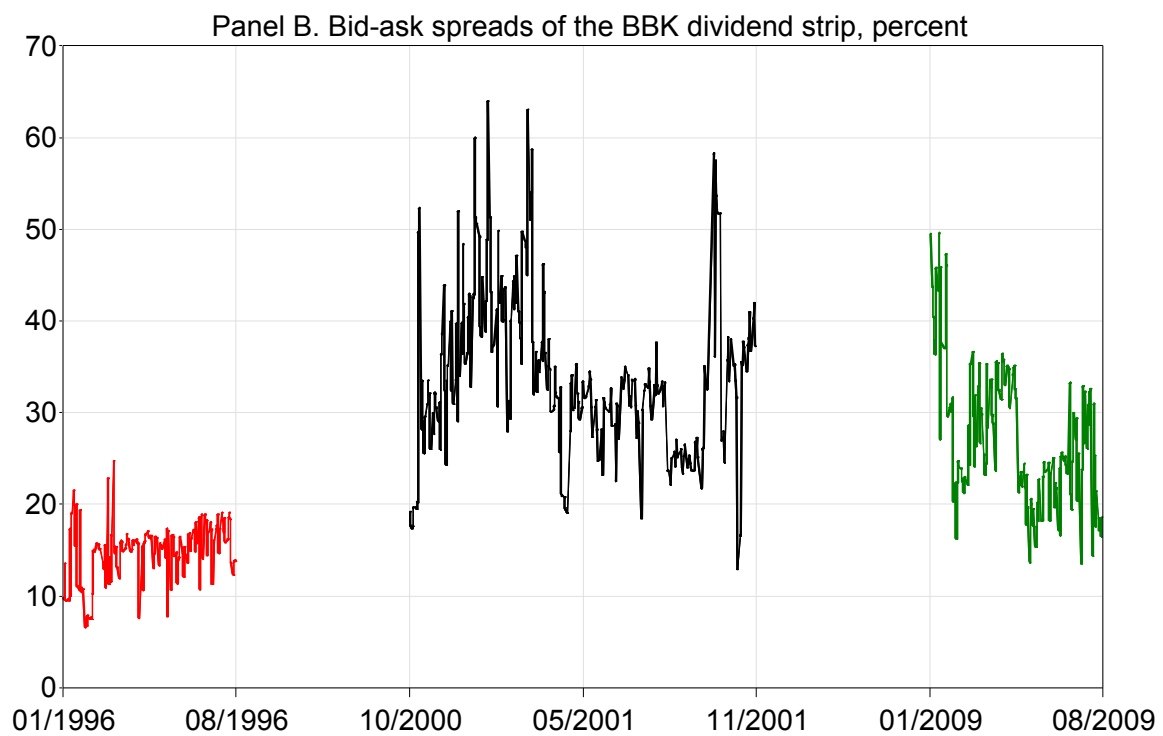
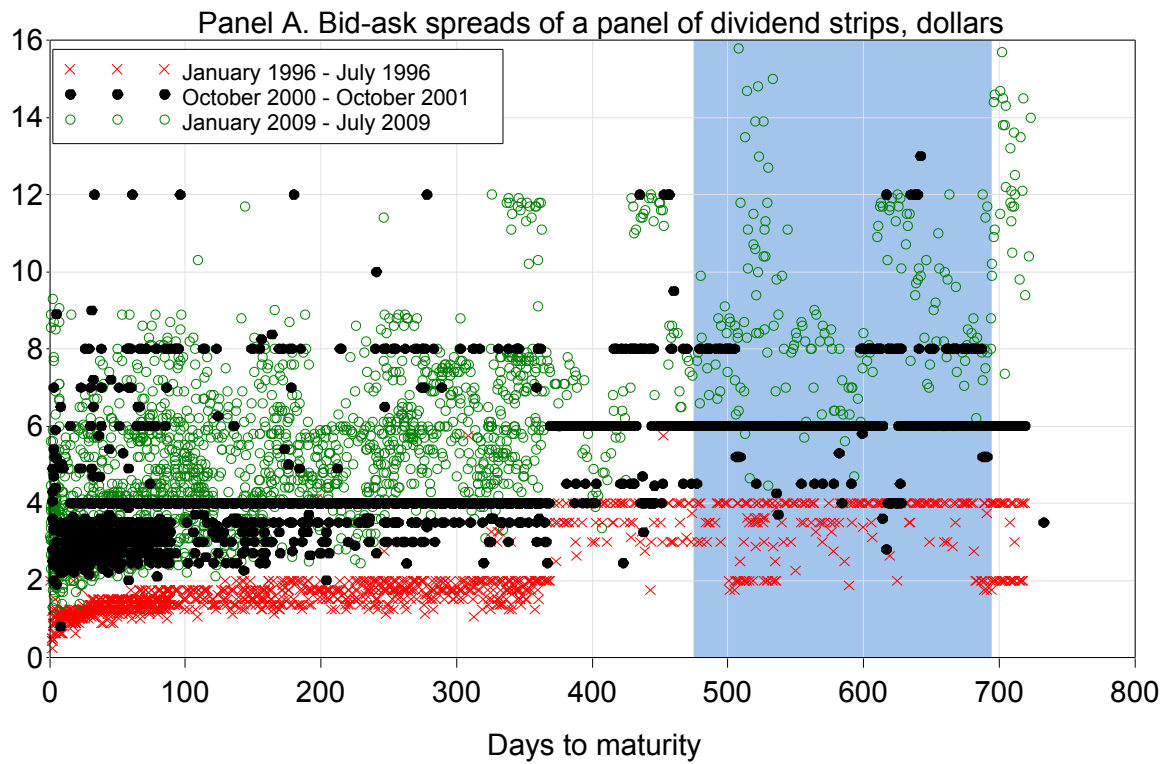


Figure 4. Bid-Ask Spreads of Short-Term Assets. This figure plots the absolute quoted bid-ask spreads of an investment in a long call and short put as a function of option maturity in Panel A. The highlighted maturities correspond to those considered by BBK. Panel B shows the spreads over time scaled by the price of the BBK short-term asset.

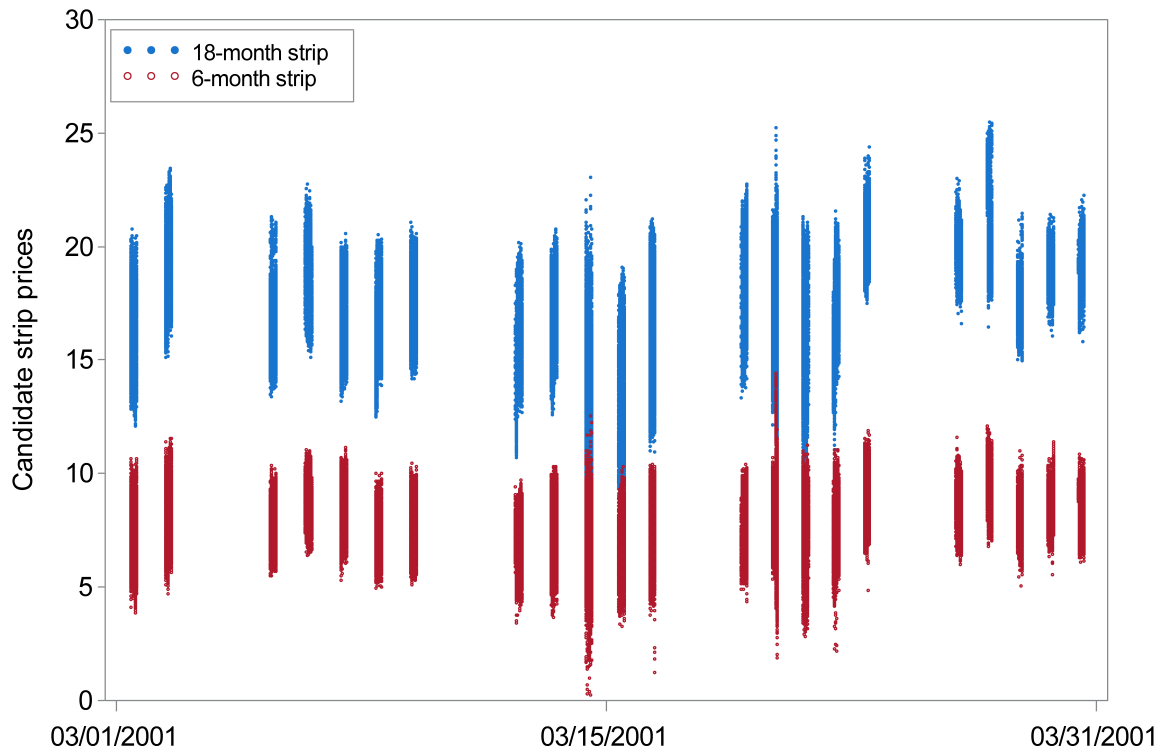


Figure 5. Candidate Short-term Asset Prices. This figure plots the prices of 18 (blue) and 6 months (red) strip candidates calculated for each matched call and put quote during March 2001.

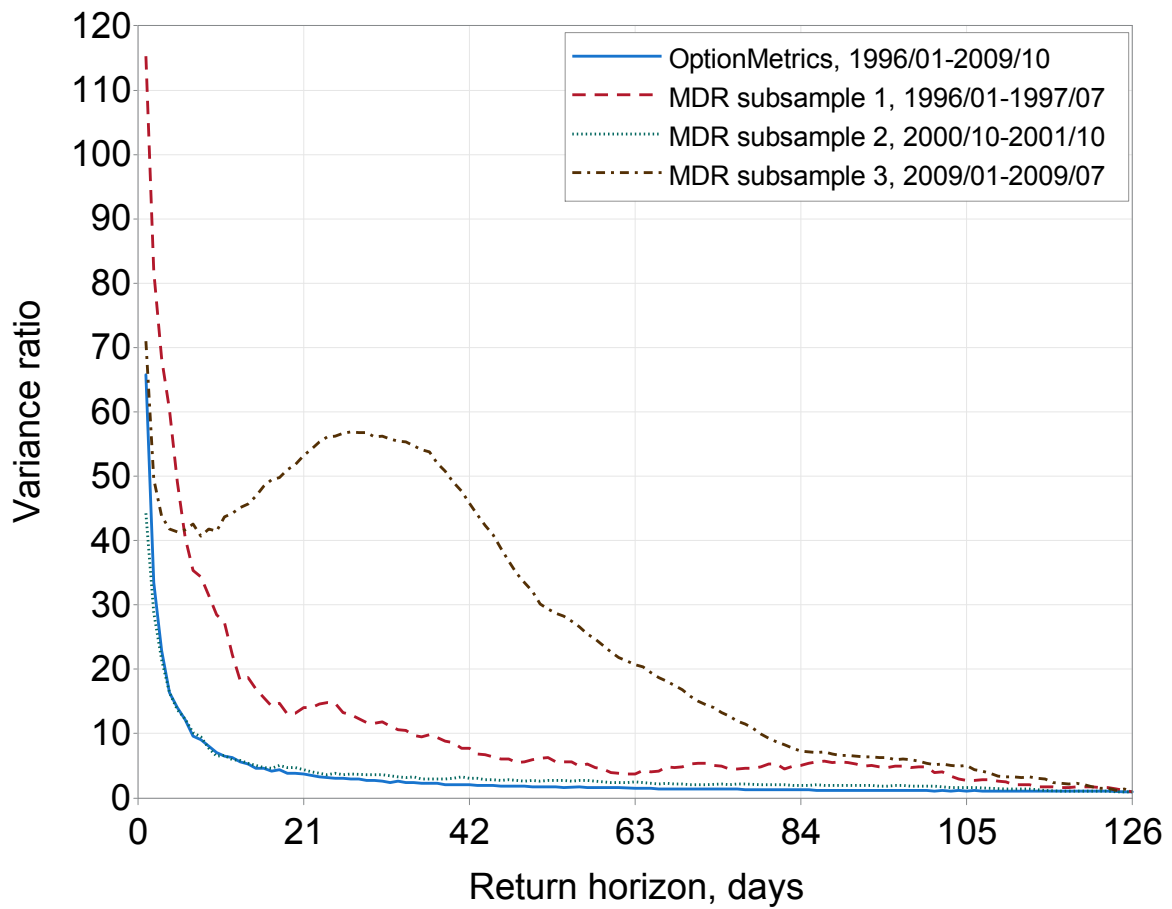


Figure 6. Variance Ratios of Short-Term Asset Returns. This figure plots variance ratios against holding horizons for investment in the short-term asset during the subsamples described in Table 1.

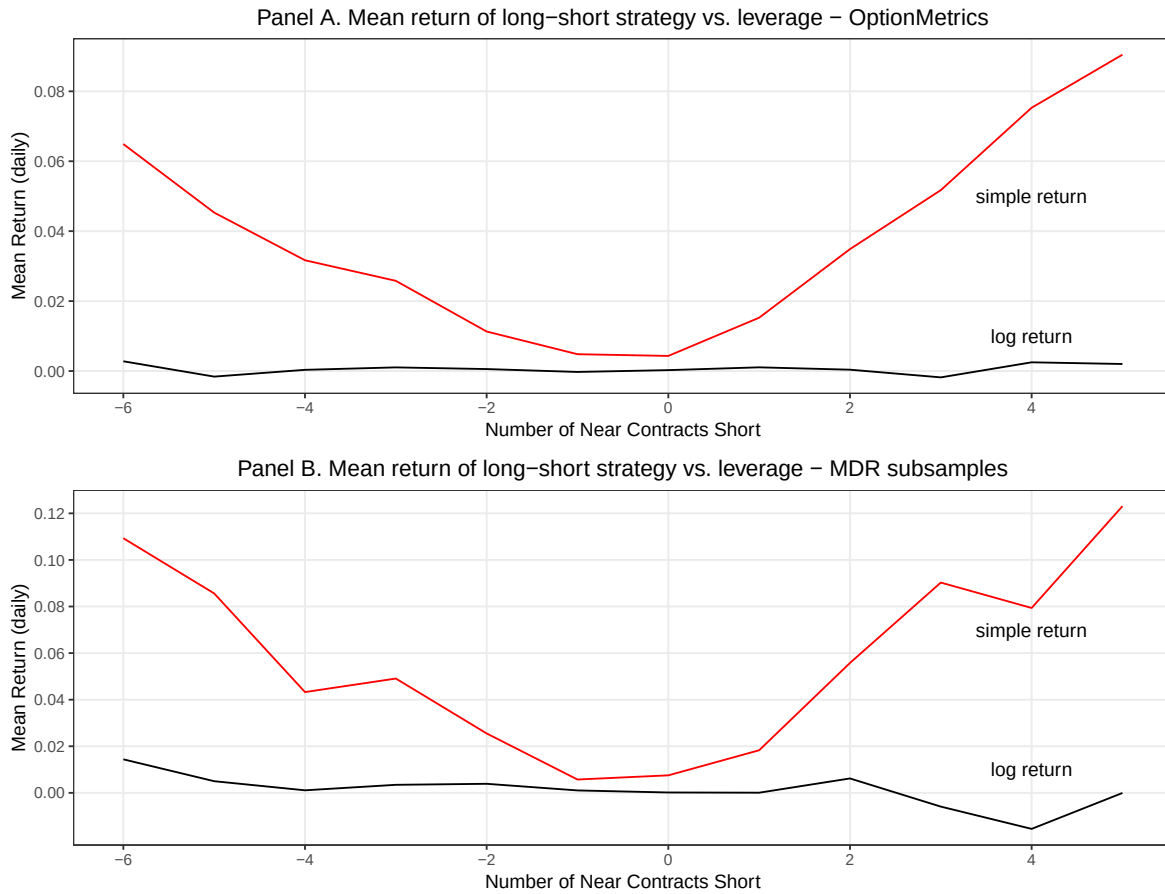


Figure 7. Mean Return vs Leverage. This figure shows how increasing leverage in options strips impacts mean returns. Two options strip strategies that produce exposure to S&P 500 dividends in six-month windows are considered: 1) the “Near” strategy dividend window at the beginning of each calendar quarter covers the time interval approximately 6-12 months in the future (reset quarterly), and 2) the “Far” strategy, with dividends in a window that is 12 months more distant than the near strategy. Panel A plots average daily returns as a function of leverage, where the x-axis indicates the number of near contracts $\mathcal{L}$ in the strategy that is $\mathcal{L} + 1$ far contracts long. The options strip returns are constructed using the OptionMetrics data covering the time period 1996:01-2016:04. Panel B plots average daily returns versus leverage for the same strategies constructed using data from the MDR subsamples as described in Table 1.

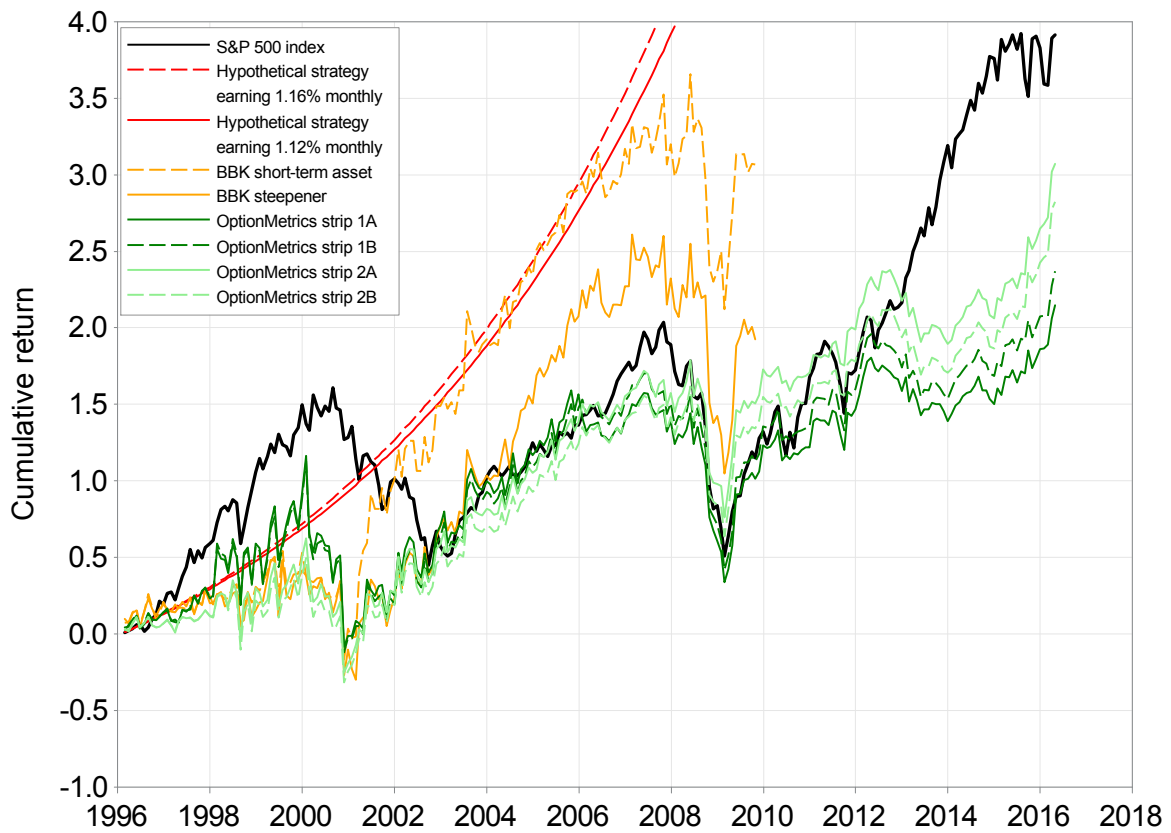


Figure 8. Comparison of Investment Performance. This figure plots the cumulative returns for the S&P total return index, the OptionMetrics end-of-day strips, the hypothetical strategies earning monthly returns of 1.16% and 1.12%, and the BBK short-term asset and steepener. Cumulative returns of the hypothetical strategies reach, respectively, 14 and 15.5 by the end of April 2016, and are truncated.